WOLAITA SODO UNIVERSITY

SCHOOL OF INFORMATICS

DEPARTMENT OF COMPUTER SCIENCE

Project Title: Web-Based Wolaita Dicha Football Club Management System With Match Prediction And Tactical Advice

A project Submitted To Department Of **Computer Science** in Partial Fulfillment for the Requirement of Bachelor of Science Degree in **Computer science**

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October 10, 2025

Declaration

Wolaita Sodo University School of Informatics Department of computer science Full name of the students 1. Tinsae Teklu 2. Melat Nurga 3. Melekot Mesfin 4. Tirusew Berihun 5. Rediet Tesfaye This is to certify that the final year web-based Wolaita Dicha Football club management system with match prediction and tactical Advice prepared by name of Tinsae Teklu, Melat Nurga, Melekot Mesfin, Tirusew Berihun, Rediet Tesfaye Titled: web-based Wolaita Dicha Football club management system with match prediction and tactical Advice for department of computer science, school of informatics Wolaita Sodo university and submitted in partial fulfillment of the requirements for the Bachelor of Science Degree in computer science complies with the regulations of the University and meets the accepted standards with respect to originality and quality.

Signed by the Examining and Advisor Committee: -

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Acknowledgement

We would like to express our sincere gratitude to all individuals and institutions who are contributing to the ongoing development of this proposal. First and foremost, our deepest thanks go to the Almighty God for granting us the strength, perseverance, and wisdom as we continue to progress through the stages of proposal writing and system development.

We are especially grateful to our project advisor **Temsegen Mengistu(Msc.)** , whose invaluable guidance, constructive criticism, and consistent encouragement.

List of Acronyms

AI Artificial Intelligence

ARMIA Auto Regressive Integrated Moving Average

BR Business Rule

CS Computer Science

ERD Entity-relationship Diagram

FC Football Club

LSTM Long Short-Term Memory

MS Master Degree

MS Medical Staff

NFR Non Functional Requirement

RNN Recurrent Neural Network

TD Technical Director

UC Use Case

UI User interface

WDFC Wolaita Dicha Football Club

WSU Wolaita Sodo University

Abstract

The advancement of digital technology and artificial intelligence (AI) has significantly transformed how organizations manage operations, analyze data, and make informed decisions. Despite this global trend, football clubs in Ethiopia, such as Wolaita Dicha Football Club, continue to rely on manual and fragmented systems for managing player data, match statistics, and tactical planning. This lack of technological integration has led to operational inefficiencies, limited real-time analysis, and minimal use of data-driven strategies—factors that hinder overall performance and competitiveness.

In response to these challenges, this project proposes the design and development of a comprehensive web-based football club management system tailored specifically for Wolaita Dicha FC. The system centralizes operations such as managing player profiles, match scheduling, training coordination, and performance tracking. A key innovation of this system is the integration of an AI agent powered by time series-based machine learning models algorithms. These models analyze sequential historical data—such as team form, player fitness, and opponent performance to predict upcoming match outcomes (win, draw, or loss) and **to** generate tactical recommendations for coaches. The tactical module interprets AI insights to suggest optimal formations, substitutions, and strategic adjustments that align with the predicted match conditions.

Additionally, the platform features a fan engagement module, enabling supporters to access AI-generated predictions, player insights, and live updates in real time. The project's methodology includes structured data collection from coaching staff, document analysis, and the application of modern web and AI technologies to ensure the platform is intelligent, secure, and user-friendly. Ultimately, the project aims to enhance strategic planning, match preparation, and club-fan interaction—contributing to the digital transformation of Ethiopian football and demonstrating the practical role of AI agents in sports management within developing regions.

keywords: Web-based system, Football club management, Match prediction, Tactical advice, Artificial Intelligence (AI).

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Chapter One

1. Introduction

1.1 Introduction and Background of the Project

Football is more than just a sport; it is a global cultural phenomenon that unites communities, shapes identities, and drives social and economic development. In Ethiopia, football holds a special place as the most widely followed and passionately celebrated sport. It plays a vital role in fostering national unity, youth engagement, and regional pride. Among the many football clubs in the country, Wolaita Dicha Sport Club (Wolaita Dicha SC**)** stands out as one of the most dynamic and community-oriented teams in southern Ethiopia.

Wolaita Dicha Sport Club was established in 2008 G.C**.** by the Wolaita Development Association (WODA) as part of its broader mission to promote social and community development through sports. The club's name "Wolaita Dicha" literally means "Wolaita Development," symbolizing its deep roots in the local community and its commitment to representing the aspirations of the Wolaita people. Before its formation, Wolaita had earlier football representations such as Damot Football Club (founded in the 1940s) and Wolaita Tussa Football Club (founded in 1995 G.C.), both of which played significant roles in nurturing local football culture but later dissolved due to organizational and financial challenges. Learning from those experiences, WODA established Wolaita Dicha SC with a sustainable structure and a clear developmental vision.

After competing successfully in the Ethiopian National League for four consecutive years, Wolaita Dicha SC achieved a historic milestone by winning the 2012 G.C. National League Championship without a single defeat, earning promotion to the Ethiopian Premier League in 2013 **G.C.** Since then, the club has remained one of the strongest sides in the league, finishing 5th in its debut season and consistently performing competitively in subsequent years. Known for its passionate fan base—the "Bees of Wolaita"—the club has brought unprecedented enthusiasm to Ethiopian football, drawing record crowds to stadiums across the country.

Beyond football success, Wolaita Dicha SC has become a source of community pride and regional identity, representing the Southern Nations, Nationalities, and Peoples' Region

(SNNPR) at the national level. Through the support of WODA, the club's mission extends beyond sports to include youth development, social mobilization, and community empowerment. Football, for Wolaita Dicha, is not just entertainment—it is a developmental tool aligned with WODA's objectives of building stronger, healthier, and more cohesive communities [1].

Despite its impressive achievements and strong community backing, Wolaita Dicha FC faces significant operational challenges due to limited technological infrastructure. Key activities such as managing player profiles, scheduling matches, recording statistics, and evaluating team tactics are still handled manually or through fragmented systems. These outdated methods hinder effective decision-making, reduce efficiency, and limit the club's ability to adapt to the growing role of data analytics in modern football.

To address these challenges, this project proposes the design and development of a comprehensive web-based football club management system specifically tailored for Wolaita Dicha FC. The system integrates core administrative and operational functions, including player management, match scheduling, training coordination, and performance monitoring. More importantly, it introduces an AI-powered module that utilizes machine learning t**o** predict match outcomes and generate tactical recommendations based on historical data and current team conditions. This will enable coaches and managers to make data-driven decisions and enhance the club's competitive strategies.

Beyond serving Wolaita Dicha FC, the project aspires to contribute to the digital transformation of Ethiopian football**.** By demonstrating the practical application of artificial intelligence, data analytics, and modern web technologies in a local context, the system can serve as a model for other football clubs seeking to modernize their management and analytical processes. Ultimately, it highlights how technology can strengthen sports institutions, improve competitiveness, and align Ethiopian football with international standards of professional sports management.

1.2 Motivation of the project

The increasing reliance on data and technology in modern football has redefined how teams prepare, perform, and progress. Across the globe, football clubs are embracing digital transformation through data analytics, artificial intelligence (AI), and automation to enhance performance both on and off the pitch. These tools help in areas like player development, match preparation, injury prevention, and tactical planning. Observing how international clubs have gained a competitive advantage through such innovations has been a major source of inspiration for this project. Introducing similar technology into Ethiopian football, especially at the club level, can revolutionize how teams operate and compete.

In the case of Wolaita Dicha Football Club, the potential for positive impact is even more significant. The club has demonstrated commitment to excellence and has access to a talented pool of players, but lacks the technological infrastructure to fully harness that potential. With an engaged fan base and motivated staff, the club provides an ideal testing ground for integrating AI and digital systems in football management. The goal is to enhance efficiency, improve results, and support data-backed decision-making processes that were previously based largely on intuition.

Furthermore, the broader motivation behind the project is to contribute to the digital transformation of Ethiopian football. By demonstrating the effectiveness of an AI-supported management system in one club, the project could inspire similar initiatives in other local teams and regions. The vision is not just to solve a problem for one club, but to introduce a scalable and adaptable model that can influence the future of football in Ethiopia. It aligns with global trends and encourages local clubs to embrace technology for sustained growth and competitive success.

1.3 Problem Statement

Although football in Ethiopia is rapidly growing in popularity and competitiveness, the management systems behind many clubs remain outdated. Wolaita Dicha Football Club, like many others, relies heavily on manual processes for tracking player performance, organizing training sessions, and managing match data. There is a noticeable absence of a centralized digital platform that can store, retrieve, and analyze critical information over time. This fragmented and static approach leads to inefficiencies in planning, limits the ability to evaluate performance trends, and slows down overall decision-making processes.

Moreover, most tactical decisions and game strategies are based largely on personal experience and coaching instinct rather than analytical evidence derived from historical data. Without systems in place to analyze opponents' strengths and weaknesses, monitor player workloads across the season, or assess

tactical effectiveness over time, coaching staff must make critical decisions without the benefit of data-driven insights. This can lead to suboptimal strategies, preventable injuries, and missed opportunities in competitive scenarios.

In particular, the absence of time series-based analysis restricts the club's ability to capture and leverage performance trends such as form fluctuations, fatigue cycles, or long-term opponent behavior. Additionally, fans are disconnected from analytical insights and lack tools to engage with team performance on a deeper level. Addressing these gaps by implementing a smart, AI-powered system—featuring a time-aware AI agent and dedicated fan interface—will create a more informed, connected, and competitive football environment.

1.4 Objectives of the Project

1.4.1 General Objective

To develop a web-based Wolaita Dicha Football club management system with match prediction and tactical Advice.

1.4.2 Specific Objectives

- To design and implement a centralized system to manage players, match schedules, and historical performance statistics.
- To collect and preprocess chronologically ordered historical match data for time series-based machine learning analysis.
- To train a predictive model using time series forecasting techniques that can forecast the result of upcoming matches (win/draw/loss) based on temporal patterns.
- To generate coaching recommendations based on evolving tactical patterns and opponent performance trends.
- To ensure data security, privacy, and role-based access control within the web-based system to protect sensitive performance and tactical data.
- To develop an interactive module for fans to access predictions and club updates.
- To implement an AI agent capable of generating real-time predictions and tactical feedback for staff and fans.

1.5 Methods of the project

This section outlines the methodological approach planned and partially implemented for designing, developing, and evaluating the proposed web-based football club management system integrated with artificial intelligence (AI). The methodology encompasses system design strategies, data collection techniques, and analytical methods that are guiding the project through its various stages. While the project is still ongoing, these approaches are being followed to ensure a systematic and effective development process. [1]

1.5.1 Design Methodology

This project adopts the Agile Software Development Model, which emphasizes iterative progress, collaboration, and adaptability throughout the development process. Agile is particularly suited for this system because it allows continuous refinement of both the web-based management platform and the AI prediction module through short, manageable cycles known as *sprints*.

Each sprint focuses on implementing a specific functional module—such as player management, match scheduling, or tactical analysis—followed by testing, user feedback, and improvement. This iterative approach ensures that the system evolves based on feedback from coaches, analysts, and stakeholders while maintaining flexibility for refining the AI algorithms used for match prediction and tactical advice.

The Agile process used in this project involves:

1. Planning and Requirement Analysis: Requirements are gathered from Wolaita Dicha FC staff through interviews and observation, and prioritized into sprint backlogs.
2. Design: For each sprint, modules such as player registration or match prediction are designed using modern web and database architecture.
3. Implementation: The system is developed incrementally using Python (Flask), HTML/CSS/JavaScript, and MySQL, integrating the AI model in later iterations.
4. Testing: Unit and integration testing follow each sprint to verify functionality, security, and prediction accuracy.
5. Review and Deployment: Completed increments are demonstrated for feedback and deployed on a test server before proceeding to the next sprint.

6. Feedback and Adaptation: Suggestions from users and the advisor are incorporated in subsequent sprints to enhance system quality.

Through this iterative and collaborative method, the project remains flexible, user-oriented, and capable of integrating advanced AI capabilities without restarting the entire cycle.

1.5.2 Data Gathering Technique

To ensure the system is aligned with the real needs of Wolaita Dicha FC, both qualitative and quantitative data collection methods will be used. These methods are designed to support the requirement analysis phase and provide the necessary input for AI model training and tactical analysis.

The planned data collection activities include:

1.5.2.1 Interview

Interviews are planned to be conducted with key personnel at Wolaita Dicha Football Club to gain a deeper understanding of existing workflows, challenges, and system needs. These interviews will help inform both system design and the development of AI features. The targeted interviewees include:

- Head Coach: To gain insights into tactical decision-making, match preparation routines, and performance evaluation methods.
- Team Analyst: To understand how performance data is currently collected, analyzed, and applied to team strategy.
- Club Administrator: To explore the current practices related to player management, data recording, storage, and match scheduling.

1.5.2.2 Observation

Direct field observations are planned to be conducted during team training sessions and pre-match preparation meetings. These sessions will offer practical insights into how the team operates in real time. The objectives of the observation phase include:

- Understanding how tactical decisions are made and communicated.
- Observing how player performance and fitness are monitored.
- Identifying how information is currently shared among coaching staff and players.

1.5.2.3 Document Analysis

As part of the requirement analysis and AI model preparation, a thorough document analysis is planned. This will involve reviewing various club records to extract relevant features for the system's database and prediction model. The documents to be analyzed include:

- Historical match reports for performance trends and outcomes.
- Tactical plans from previous seasons to understand team strategies.
- Player performance logs detailing individual statistics.

1.5.3 Data Preprocessing Technique

Before developing the prediction and tactical recommendation modules, a comprehensive data preprocessing phase was conducted to ensure that the dataset is clean, consistent, and ready for machine learning using the XGBoost Classification Algorithm (XGBClassifier). The dataset consists of historical match records, player performance data, and contextual information such as match location and injury status. The preprocessing process was crucial for transforming these raw records into a structured format suitable for supervised classification tasks.

The main preprocessing steps include:

- **Data Cleaning:**
  Missing values, inconsistencies, and outliers were identified and handled to maintain the accuracy and integrity of the dataset. Duplicate records were removed, data types were standardized, and incomplete entries were treated using statistical imputation or removal depending on their importance.
- **Feature Extraction and Transformation:**
  Key attributes such as *Goals For*, *Goals Against*, *Shots*, *Possession*, *Home/Away*, *Number of Injured Players*, and *Key Player Injury Status* were selected as predictors. Additional calculated features, including *Wins_last5*, *Draws_last5*, and *Losses_last5*, were derived using the add_recent_form_features() function to represent team momentum and recent performance trends.
- **Numerical Normalization:**
  Continuous numeric values, such as goals scored, shots, and possession percentages,

were normalized to comparable ranges to prevent features with large numerical magnitudes from dominating the model's learning process.

- **Feature Aggregation:**
  Aggregate indicators such as recent averages and rolling performance metrics were computed using the mean() function. These features help capture form stability and overall team consistency.
- **Dataset Splitting:**
  The preprocessed dataset was divided into training and validation subsets. The training data was used to fit the model, while the validation data was reserved for evaluating the model's accuracy and generalization.

1.5.3.1 Encoding Techniques

To make categorical and binary data understandable by the machine learning model, two main encoding techniques were applied:

- **Label Encoding:**
  The match outcome labels ("Win", "Draw", "Loss") were transformed into numeric representations (e.g., 0, 1, 2) using the LabelEncoder() method. This encoding enabled the XGBoost model to interpret categorical outcomes during supervised learning.
- **Base64 Encoding:**
  Base64 encoding was applied to encode binary data such as images, logos, and secure tokens into text-based representations. This method allows safe data transmission and storage within the web system without losing information integrity.

These encoding processes ensure that both categorical and binary information are efficiently processed by the prediction engine and the Flask-based web interface, supporting accurate model training, evaluation, and deployment.

1.5.4 Model Development Technique

The machine learning model is not yet developed, but we have outlined a plan for its design and training using time series-based approaches. The model will be used to predict match outcomes and generate tactical suggestions by analyzing trends and patterns over time from sequential match data. The following steps are proposed:

Algorithm Selection :

The predictive model in this system is developed using the XGBoost Classification Algorithm (XGBClassifier), an ensemble learning technique based on the principle of Gradient Boosting. XGBoost constructs many small decision trees sequentially, with each tree learning from the errors of the previous ones. The final prediction is a weighted combination of all trees, resulting in high accuracy and strong generalization.

Compared with classical time-series models such as LSTM, RNN, and ARIMA, XGBoost provides faster computation, efficient handling of mixed numerical and categorical features, and excellent performance in multiclass classification problems such as predicting match outcomes (Win / Draw / Loss).

Supporting algorithms and functions include:

- LabelEncoder(): Converts categorical results (“W”, “D”, “L”) into numeric labels.
- add_recent_form_features(): Computes team form from the last five matches.
- Mean(): Calculates rolling averages for recent performance
- predict_proba(): Generates probability distributions for each possible outcome.
- tactical_advice(): Applies rule-based logic to translate numeric outputs into human-readable tactical suggestions.

This hybrid structure enables both high-accuracy prediction and interpretable tactical recommendations for coaches and analysts.

- Time-Aware Model Training: Once the dataset is chronologically structured, the selected algorithms will be trained to recognize temporal patterns in features such as recent match outcomes, player fitness trends, team formations, and opponent strengths. The models

will be trained using time-based cross-validation or walk-forward validation to ensure realistic forecasting.

- Hyperparameter Tuning: We will apply optimization techniques such as Grid Search and Random Search to tune model parameters, specifically for time series configurations (e.g., number of time steps, sequence length, dropout rates in LSTM, ARIMA order parameters).
- Incorporating Tactical Logic: The prediction model will be complemented by tactical logic modules or decision-rule engines. These will analyze time series outputs (e.g., declining performance trends or repeated tactical failures) and generate dynamic strategy recommendations tailored to upcoming opponents and player conditions.

1.5.5 Model Evaluation Technique

The predictive model is evaluated using the **Multiclass Logarithmic Loss (Log Loss / Cross-Entropy Loss)** metric during training. Log Loss measures the accuracy of predicted probability values rather than simple categorical labels, penalizing confident but incorrect predictions more heavily. Lower Log Loss values indicate better performance.

The model is trained with parameters such as:

```
model = XGBClassifier(

n_estimators=200,

use_label_encoder=False,

eval_metric='mlogloss',

random_state=42

)
```

During evaluation, the system monitors Log Loss on validation data to ensure optimal generalization across match conditions. Once deployed, real-world match outcomes will be compared with predicted probabilities to continuously assess and refine model accuracy.

1.5.6 Model Attributes (Features Used for Training)

The model is trained using ten core attributes derived from historical and contextual match data. Each attribute contributes unique predictive insight into the team's performance and outcome likelihood.

Table 1:Model attribute

Feature	Description	Source
Goals For	Total goals scored by Wolaita Dicha FC in a match	Dataset
Goals Against	Total goals conceded	Dataset
Shots	Number of shot attempts	Dataset
Possession	Team possession percentage	Dataset
Home	1 if Wolaita Dicha is home, 0 otherwise	Dataset
Num_Injured	Total number of injured players	Dataset
Key_Player_Injured	1 if a key player is injured	Dataset
Wins_last5	Wins recorded in the last five matches	Calculated
Draws_last5	Draws recorded in the last five matches	Calculated
Losses_last5	Losses recorded in the last five matches	Calculated

These combined features capture raw match statistics, contextual conditions, and recent form trends, allowing the XGBoost model to learn complex relationships that influence match results.

1.5.7 Model Deployment Technique

After model training and evaluation, the deployment phase integrates the prediction engine with the web-based football club management system. The deployment process ensures that the trained XGBoost Classification Model (XGBClassifier) operates seamlessly within the Flask web framework, enabling real-time interaction between users and the AI module. The deployment technique involves the following key steps:

- **Model Serialization:**
  Once the XGBoost model achieves the desired accuracy, it is serialized and saved using Python's Joblib or Pickle library. This enables efficient reuse of the trained model without retraining every time the system is launched.
- **Backend Integration (API Development):**
  A RESTful API is implemented using the **Flask** framework. This API acts as the bridge between the prediction model and the web interface. When users submit new match data (e.g., player fitness, previous results, or opponent information), the API sends the data to the model, retrieves the predicted outcome, and returns it to the web dashboard.
- **Web Application Integration:**
  The serialized model is fully integrated into the Flask backend. The system's interface allows club staff to input data and receive AI-powered match predictions and tactical advice instantly. The integration ensures smooth communication between the frontend, backend, and database layers.
- **Hosting and Deployment Environment:**
  For development and testing, the complete web application—including the AI model and database—is hosted locally using XAMPP or Flask's built-in server.
  For live online access, the project can be deployed on **free** hosting platforms. These services provide reliable, no-cost hosting options suitable for academic and prototype systems.
- **Continuous Monitoring and Model Updates:**
  After deployment, the system's performance and model accuracy are continuously monitored. As new match data becomes available, the AI model will be **retrained and**

re-serialized periodically to ensure up-to-date tactical recommendations and accurate match predictions.

This deployment approach guarantees that the system remains **accessible, maintainable, and cost-effective**, while providing a scalable foundation for future upgrades and cloud migration if needed.

1.6 Scope of the project

The scope of this project focuses on the development of a web-based football club management system for Wolaita Dicha Football Club, enhanced with an integrated AI agent that leverages time series-based machine learning to predict match outcomes and provide tactical guidance. The system also aims to actively involve fans by offering access to performance insights, match forecasts, and team updates via a dedicated fan interface. The main features include:

- Player Management: Stores personal details, injury history, playing position, and performance data over time.
- Match Management: Records historical match details (date, opponent, result, key statistics) and supports future fixture scheduling.
- Data Analytics Dashboard: Provides visual trends of player performance, team form, and statistical comparisons across timelines.
- AI Prediction Engine: Uses chronologically structured match data to forecast the outcomes of upcoming games (win/draw/loss) based on temporal trends and opponent dynamics.
- Tactical Recommendation System: Offers automated, AI-driven strategic suggestions for coaches, considering recent performance patterns and opposition analysis.
- Fan Engagement Module: Allows fans to view match predictions, player stats, and summaries generated by the AI agent, improving transparency and excitement.
- User Roles: Secure, role-based access for administrators, coaches, analysts, players, and fans, with tailored interfaces and permissions.

1.7 Significance of the Project

The development of a web-based football club management system integrated with artificial intelligence will hold significant value for various stakeholders involved in the day-to-day operations and strategic planning of Wolaita Dicha Football Club. Once developed, the system will increase efficiency and organization while introducing data-driven decision-making into a traditionally manual environment. The significance of this project can be assessed from the perspective of different user groups:

1.7.1 For Customers (Fans)

- Enhanced Fan Experience: The inclusion of a fan-focused module allows supporters to engage more deeply with the club through access to real-time match predictions, performance insights, and team updates powered by the AI agent. This improves fan satisfaction and loyalty by making them part of the club's analytical ecosystem.
- Improved Team Performance: Through AI-driven match preparation and tactical analysis based on time series modeling, the team is expected to perform more consistently, increasing fan enjoyment and pride.
- Greater Professionalism and Transparency: By digitizing key operations and making AI-generated insights accessible to fans, the club boosts its image as a modern, data-driven organization aligned with international football standards

1.7.2 For Employees (Coaches, Players, Technical & Medical Staff)

- Data Access and Insights: Once the system is developed, coaches will be able to review player statistics and match outcomes in a centralized platform, and players will have access to their performance history and training records.
- Informed Decision-Making: AI-powered suggestions will help coaching staff make smarter tactical decisions, reduce guesswork, and identify performance gaps.
- Efficiency: By eliminating manual processes, such as filling out paper logs or searching for old records, the system will save time and reduce errors.

1.7.3 For the Association (Club Management and Governing Bodies)

- Strategic Planning: The system will offer data that can be used to assess coaching performance, monitor long-term player development, and support transfer decisions once it is implemented.

- Financial Efficiency: By tracking injuries, and player usage, the system will contribute to better resource management in the future.
- Compliance and Reporting: A centralized database will simplify regulatory reporting and documentation for league authorities.
- Technology Adoption: This project will position the club as a technological pioneer within Ethiopian football, potentially attracting investment and partnerships once the system is operational.

1.8 Feasibility Study

A feasibility study evaluates the practicality and viability of a proposed system before its full implementation. It ensures that the project can be realistically achieved using the available resources, technologies, and organizational support. The feasibility of the Wolaita Dicha Football Club Management System has been assessed across four key dimensions: operational, economic, technical, and legal feasibility. Although the system is still under development, early assessments suggest promising potential.

1.8.1 Operational Feasibility

This section evaluates whether the proposed system can be smoothly integrated into the club's daily operations.

- **User Readiness:** Initial discussions with coaches, administrators, and analysts at Wolaita Dicha FC show strong enthusiasm for adopting digital solutions to improve their work processes.
- **Need Justification**: The club currently faces challenges due to the lack of a centralized system for player monitoring, data management, and match planning, leading to inefficiencies and delays.
- **Organizational Alignment**: The system's goals align well with the club's mission to enhance professionalism and boost team performance, making its acceptance more likely.
- **Ease of Use**: Designed as an intuitive web-based platform, the system will be easy to learn and operate, requiring minimal training for staff.

1.8.2 Economic Feasibility

This evaluates whether the system is cost-effective in relation to its expected benefits.

- Low Development Cost: The use of open-source tools and technologies—such as Python, Django and Scikit-learn—minimizes licensing fees and overall development expenses.

- Minimal Hardware Requirements: As a browser-accessible web application, the system requires no major hardware upgrades; existing laptops and internet access are sufficient.
- Return on Investment (ROI): Once implemented, the system is expected to reduce administrative overhead, support better match preparation, and improve decision making—potentially leading to stronger performance and greater fan engagement or sponsorship opportunities.
- Low Maintenance Cost: Ongoing maintenance is expected to be minimal, involving periodic software updates and minor bug fixes.

1.8.3 Technical Feasibility

This examines the technical resources and capabilities needed to implement and maintain the system.

- Technology Stack: The system is being developed using proven technologies, including Python for backend logic and AI modules, JavaScript for interactivity, and MySQL for secure data management.
- Development Expertise: The required technical expertise—covering web development, database design, and machine learning—is available within the development team. Frameworks like Django support rapid prototyping and iteration.
- Infrastructure: The system will be deployed on a cloud-based or local server and will be accessible from multiple devices through a standard web browser.
- Scalability: The system is being designed with scalability in mind, allowing for future enhancements such as mobile app integration, real-time data updates, or extension to support multiple clubs.

1.8.4 Legal Feasibility

This assesses compliance with relevant legal, ethical, and data protection standards.

- Data Privacy: Sensitive information such as player statistics, medical records, and performance data will be securely stored and restricted to authorized users.

- Data Usage Rights: All data intended for use in system development and training, including match performance data, originates from the club's internal records, ensuring ethical and lawful usage.
- Regulatory Compliance: The system is designed to operate within the policies and regulations of the Ethiopian Football Federation and adheres to applicable data protection principles governing internal systems.

1.9 System Requirements

This section outlines the **hardware**, **software**, and **programming tools** will require to develop, deploy, and run the proposed Wolaita Dicha Football Club Management System. These requirements are based on the scope of the project and its expected performance needs.

1.9.1 Hardware Requirements

The system is designed to run efficiently on standard computing infrastructure without the need for high-end hardware. However, to ensure smooth operation for both development and usage.

1.9.2 Software Requirements

The following software tools are required for development, deployment, and operation of the system:

A. Development Environment

- **Backend Framework:** Python with Flask serves as the core backend framework responsible for routing, RESTful API integration, and real-time communication between the web interface and the machine-learning model.
- **Frontend:** HTML5, CSS3, JavaScript (with Bootstrap for responsive UI)
- **Database:** MySQL
- **Machine Learning Libraries:** Scikit-learn like Label Encoder.
- **IDE:** VS Code

B. Deployment Environment

- **Web Server:** Apache
- **Hosting Platform:** Xampp

1.9.3 Programming Language

The system uses modern and widely supported programming languages for both web and AI development:

Table 2:Programming Language

Component	Programming Language
Backend	Python and php
Frontend	HTML,CSS, JavaScript
Machine Learning Model	Python (Scikit-learn)
Database Interaction	SQL (MySQL)

Chapter Two

2. Description of the Existing System

2.1 Introduction of the Existing System

In this chapter we studied the existing system deeply, since it is necessary to know the existing working system of office so as to develop a better system. When we studied the existing system we gave emphasis for here under listed questions:

- How the existing system is working?
- What kind of method they use to stored data?
- What is the business rule they use?

After studying the existing system, we also determined the requirement or the feature that must be included in the proposed system. Furthermore by analyzing the current system, we could also estimate how the propose system solve the setbacks of the existing system. This chapter formally defines the detailed functional user requirements using high-level requirements identified in the initiation, System Concept, and planning phases. It also defines the requirements in terms of data, system performance, security, scalability and maintainability requirements for this project. The requirements defined in this part to a level of detail sufficient for systems design to proceed. They need to be measurable, testable, and relate to the business need or opportunity identified in the initiation Phase. This chapter in general focuses on overview of the existing system, overview of the proposed system, functional and non-functional requirements of the proposed system and proposed solution for the new system to address problems of the existing system.

The existing management system of the Wolaita Dicha Football Club is primarily manual and paper-based, with limited integration of modern digital tools. Most administrative activities, including player registration, match scheduling, financial tracking, and staff coordination, are handled through physical records and traditional documentation methods. This leads to several challenges such as data inaccuracy, difficulty in retrieving information, inefficient communication among departments, and delayed decision-making processes.

Moreover, the lack of a centralized database makes it hard to manage player performance statistics, medical records, and training schedules efficiently. Stakeholder engagement, including fan interaction and sponsor communication, is minimal due to the absence of an online platform. Overall, the current system does not meet the growing operational demands of a professional

football club, highlighting the need for a robust, automated, and integrated management solution. Currently, the management and tactical operations of Wolaita Dicha Football Club are handled using traditional/manual methods combined with basic digital tools.

At present, Wolaita Dicha Football Club does not have an integrated computerized management and tactical support system. Instead, the club relies on a combination of manual processes, basic digital tools, and human expertise to manage its operations.

2.2 Player in the Existing System

The existing management system of Wolaita Sodo Football Club is largely manual and fragmented, involving several core stakeholders who perform essential functions. These players include:

Technical Director

- Oversees all football-related operations within the club. This includes full control over player data, performance tracking, player status (injuries, transfers, etc.), and managing the coaching staff. The technical director is responsible for making informed decisions based on player development and match analysis. However, under the current manual system, gathering and analyzing performance data is difficult and inefficient, leading to delays in decision-making and limited strategic planning.

Coach

- Focuses on conducting training sessions, developing match strategies, and guiding players. Coaches depend on the technical director for player-related data and status. Without a digital system, coordination between coaching staff and management is slow and lacks real-time insights.

Players:

- The club's athletes, whose records such as personal information, training performance, and injury history are stored in physical files. This makes data management vulnerable to errors and delays, affecting player development and monitoring.

Fans:

- Supporters of the club who currently have limited interaction with team activities due to the lack of an official online presence. Information related to matches, news, and merchandise is not easily accessible, which limits fan engagement.

Medical Staff:

- The medical staff manages player health information manually. Injury reports, treatment details, and fitness updates are recorded on paper or stored in separate files without a centralized database. Communication between the medical team, coaches, and management is often done through phone calls or handwritten notes, which makes it difficult to keep accurate and updated health records.

2.3 Function / Activity of the Existing System

The current system used by Wolaita Dicha Football Club is mostly manual and paper-based, with limited automation or centralized data handling. The major functions and activities carried out within this existing system include:

Player Registration and Record Keeping:

- Personal details, contact information, and health status of players are recorded manually.
- Files are stored physically, making it difficult to update, retrieve, or manage efficiently.

Match Scheduling:

- Matches are planned using handwritten schedules or basic digital tools (like spreadsheets).
- Receiving fixtures from the football federation and entering them manually into spreadsheets or notice boards.

Performance Tracking:

- Player performance is tracked using physical reports, with no automated analytics or performance dashboards.
- Coaches and the technical director manually compile and analyze data, often leading to inconsistent evaluations.

Injury and Medical Record Management:

- Medical updates and injury reports are documented on paper.
- There is no centralized system for doctors and coaches to access a player's health history quickly.

Coaching and Training Management:

- Training plans are designed and delivered without any digital support.
- Monitoring progress is difficult due to the absence of structured, digital performance logs.

Administration and Documentation:

- All club-related documents are handled manually, including contracts, player transfers, and internal communications.
- This creates a risk of data loss, redundancy, and inefficiencies in processing.

Fan Engagement:

- Communication with fans is limited to local announcements, word-of-mouth, or informal social media use.
- No official platform exists for fans to register and support the team digitally.

Marketing and Sponsorship Management:

- Sponsorship deals and marketing activities are managed using basic documentation.
- Tracking sponsor engagements and marketing outcomes is difficult without structured reporting tools.

2.4 Business Rule

A business rule of existing system is successfully an operating standard or policies that will explain for existing system of Wolaita Dicha Football Club Management System.
The existing system has many business rules or principles some of them are:
BR1: New players must be registered manually using paper forms or spreadsheets by the administrative staff.

BR2: Only authorized staff (administrators, coaches, or medical staff) can access player records.

BR3: Player injuries must be reported immediately to the medical staff and recorded manually.

BR4: Training attendance must be marked daily by coaches on paper lists or sheets.

BR5: Match fixtures must be received from the football federation and posted manually on notice boards or spreadsheets.

BR6: Official match reports must be prepared manually after each game by the technical team.

BR7: Tactical formations and strategies must be designed and approved by the head coach only.

BR8: Opponent analysis should be done through video replays or scouts' reports, not automated systems.

BR9: All expenses and payments must be documented in physical ledgers or spreadsheets, and approved by club administrators.

BR10: Medical staff must report injuries and health status to the technical director using paper reports.

2.5 Bottlenecks of the Existing System

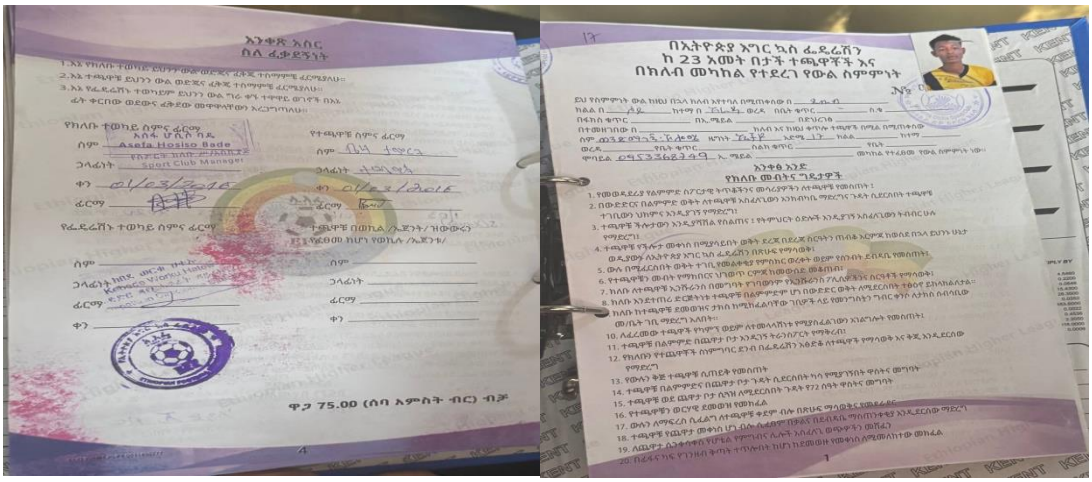
In Wolaita Dicha Football Club, management of players, matches, and tactical decisions is applied in a manual way of implementing tasks. The existing system faces many technological and organizational problems. Some of the major bottlenecks of the manual system are:

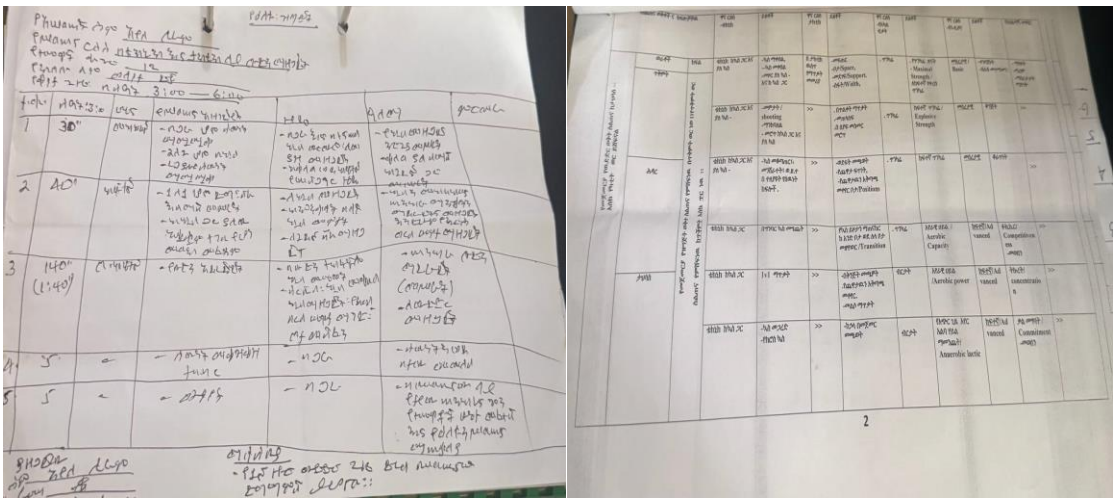
- Difficult to get real-time information about player performance, injuries, or training attendance, which affects timely decision-making.
- Difficult to manage the overall system since records are scattered in paper files and spreadsheets.
- Player file and contract control mechanism is tedious and complicated to update and retrieve.
- High risk of data redundancy and inconsistency due to lack of a centralized database.
- Difficult communication between management, coaches, players, and fans since most communication is verbal or via informal channels.
- When a new player joins or transfers without proper documentation, only limited information is stored, making it hard to track history.

2.6 Forms of the Existing System

The current data management practice at Wolaita Dicha Football **Club** relies on traditional, manual approaches that are not integrated or automated. Information related to players, matches, training, and administrative activities is recorded in physical documents such as notebooks, printed reports, and paper files. Some data, such as match results or player lists, may also be stored in isolated digital files like Microsoft Excel spreadsheets. However, these files are not centralized, making it difficult to update, share, or analyze information efficiently.

To better illustrate the current workflow, the following figure shows the **existing system data storage and management process**, where each department maintains its own unconnected data store without centralized access or real-time communication.





Chapter Three

3. Proposed System Modeling

3.1 Proposed System Requirement Specification

In the world of technology and engineering, System Requirements are essentially the conditions or capabilities that a system must meet. They define what the system is expected to do and how well it should perform. Think of them as the blueprint and the performance targets for building or acquiring a system.

To reduce complexity of existing system, effective management of time, to make work easy, simple and error free, effective utilization of available resource, to enhance the efficiency and, to design user friendly and interactive graphical user interface; requirement analysis is the key phase in project development. Requirement divided into two: function requirement and nonfunctional requirements.

The proposed system for Wolaita Dicha Football Club aims to overcome the limitations of the existing manual system by introducing a computerized, data-driven management platform. The system will support player management, match scheduling, performance analysis, match prediction, and tactical advice generation.

3.1.1 Functional Requirement

Functional requirements specify what a system must do, outlining the behaviors, functions, and tasks it needs to perform to meet user needs, such as creating a user profile or processing a payment. These requirements detail the relationship between system inputs and outputs, defining the specific actions the system takes, and are documented through methods like written descriptions or use cases.

The following are the functional requirement that the system must meet:-

- The system shall allow registration, updating, and retrieval of player information (personal details, contracts, positions, fitness, and performance).
- The system shall maintain player training attendance and history in a centralized database.
- The system shall record match fixtures, venues, results, and statistics.

- The system shall analyze player and opponent data to suggest optimal formations and strategies.
- The system shall provide coaches with real-time tactical recommendations before and during matches.
- The system shall use historical data, player form, and opponent statistics to predict the probability of match outcomes (win, draw, loss).
- The system shall display predictions in percentage form for better analysis.
- The system shall allow medical staff to record injuries, treatments, and recovery status of players.
- The system shall link fitness records with performance data to monitor progress.
- The system shall send notifications to authorized users for upcoming matches, training sessions, or critical updates.

3.1.2 Non-Functional Requirement

Non-functional requirements place constraints on how the system will do so. The non-functional requirement elaborates a performance characteristic of the system. Also these requirements relate to system attributes such as reliability and response time and they can arise due to user requirements. Any requirement which specifies how the system performs a certain function considered when designing the solution.

The following are non-functional requirement associated with the system:-

NFR1: Performance

- The system shall provide real-time access to player and match data with minimal delay.

NFR2: Security

- The system shall restrict access to sensitive data through role-based authentication (Admin, Coach, Medical Staff, and Player).
- Data shall be stored securely to prevent unauthorized access or modification.

NFR3: Reliability

- The system shall ensure data consistency across all modules (player, match, medical, tactical).
- Backup mechanisms shall be in place to prevent data loss.

NFR4: Usability

- The system shall have a user-friendly interface that can be easily used by administrators, coaches, and players.
- The system shall support both English and local language (if required by the club).

NFR5: Scalability

- The system shall be able to handle an increasing number of players, matches, and seasons without performance degradation.

NFR6: Availability

- The system shall be available at all times except during scheduled maintenance.

3.1.3 Actor and Use case Identification

Actor: An actor represents a type of users of the system or external system that plays a role in one or more interactions with our system.

Use case: A use case describes a sequence of actions that provide a measurable value to an actor. A use case is drawn as a horizontal ellipse on a UML use case diagram.

3.1.4 Actor Identification

Actors are external factors that interact with the system. This may include Coach, Players, Technical Director, Administrator, Match Analyst, Secretary, AI and Medical Staff.

Name: -Administrator

Description: -a person who owns the system and can manage the whole system including Wolaita Dicha FC Management System.

- Manages user accounts, system settings, and overall data.
- Handles player registrations, contract details, and transfers.

Name: - Coach

- Uses tactical advice and match prediction tools.
- Reviews player performance reports.
- Manages training schedules and attendance.

Name: - Technical Director

- Oversees all football-related operations within the club. This includes full control over player data, performance tracking, player status (injuries, transfers, etc.), and managing the coaching staff.

Name: - Medical Staff

- Records injuries, treatments, and recovery progress.
- Provides fitness reports linked with player performance.

Name: - Secretary

- Inputs player information's.
- Update player detail information's.
- Register matches that comes from Ethiopian football federations

Name: - Fan

- Views match, results, and club updates.
- Register for member

Name: - AI Model

- Processes player, match, and opponent data to predict match outcomes.
- Generates tactical advice (formation, substitutions, and strategies).
- Provides automated insights to coaches and analysts.

3.1.4 Use case Identification

Use case identification is the process of identifying the interactions between **actors** (users or external systems) and the system in order to capture functional requirements.

Table 3:Use case identification

No	UC No	Use cases
01	UC01	Create user account
02	UC02	Update user account
03	UC03	Search user account
04	UC04	Activate user account
05	UC05	Deactivate user account
06	UC06	Forget password
07	UC07	Change password
08	UC08	Update profiles
09	UC09	Login
10	UC10	Register players info
11	UC11	Update player info
12	UC12	Search player info
13	UC13	Register matches (Upcoming and Recent)
14	UC14	Update matches (Upcoming and Recent)
15	UC15	Search matches (Upcoming and Recent)
16	UC16	Register results
17	UC17	Update results
18	UC18	Register for member
19	UC19	View matches and results
20	UC20	Add training schedule
21	UC21	Update training schedule
22	UC22	Approve player registration
23	UC23	Register player injury
24	UC24	Update player injury
25	UC25	Search player injury

26	UC26	Train model
27	UC27	Predicate match
28	UC28	Give tactical suggestion

3.2 Use case Diagrams

A use case diagram is a graphic depiction of the interactions among the elements of a system. A use case is a methodology used in system analysis to identify, clarify, and organize system requirements. Use Case diagrams show the various activities the users can perform on the system. The system is something that performs a function. A use case diagram illustrates a set of use cases for a system, the actors of these use cases, the relations between the actors and these use cases, and the relations among the use cases.

Use Case Diagram for WDFC Management System with Predication and Match Analysis

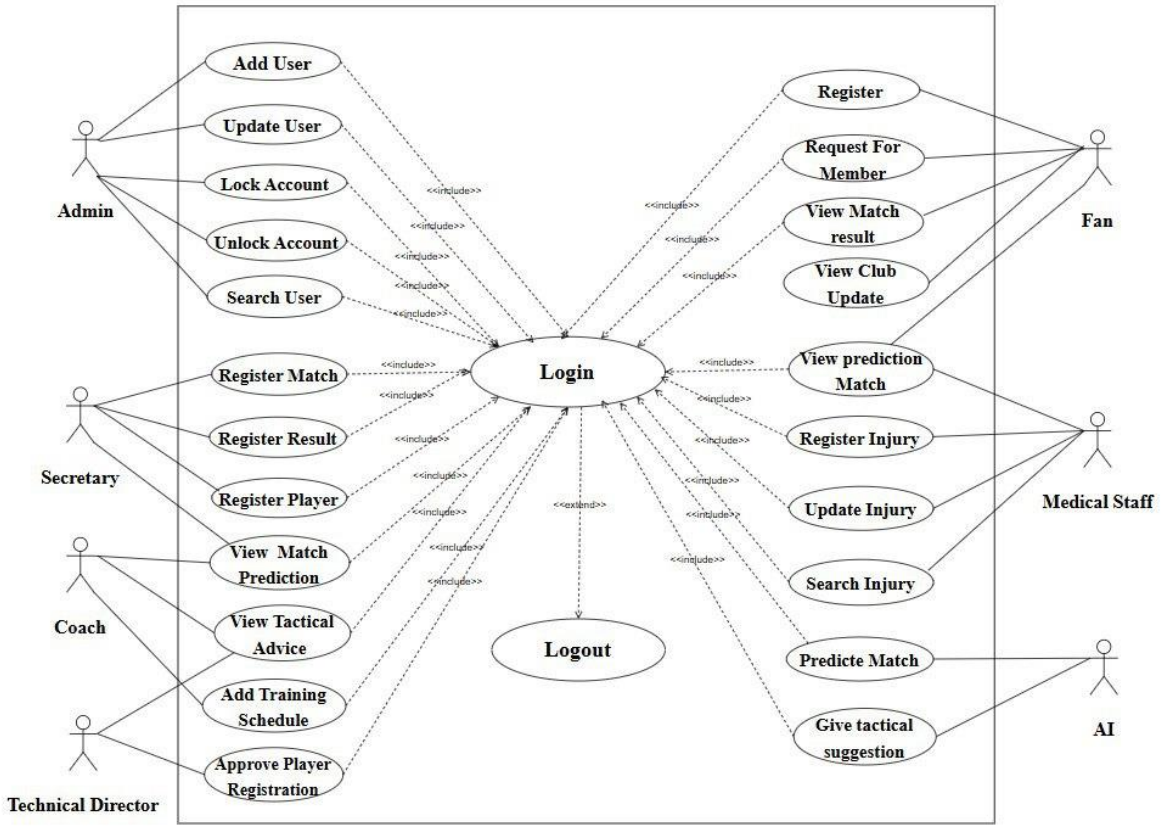


Figure 1:use case diagram

3.2.1 Use case Description

Use Case Documentation describes each use case in detail: its purpose, flow, actors, inputs, outputs, and conditions.

Add User Use case Description

Table 4:Add User Use case Description

Use Case Name	Create Account
Actor	Administrator
Pre-Condition	The actors should be an administrator The actor should be here own email or password.
Description	This state allows these actors, to login Email and password in order to access the system.
Basic course of action	1. The Administrator enter to home page 2. Then click login page 3. The Administrator fill valid the required data on the form and click login button 4. The Administrator click register link and displays registration form. 5. The Administrator enter the form field and click register button 6. The system can register on Database and display success message. 7. Use Case End
Alternate flow of action	A1. If the entered required information is not valid it returns step 4 to refill the form.
Post Condition	The new user is registered.

Login Use case Description

Table 5:Login Use case Description

Use Case Name	Login
Description	Users/Actors use this use case to log into the system.
Actors	Coach, Fan , TD, MS, Secretary and Administrator
Pre-condition	The user should be registered first or need to have account to login
Basic course of action	1. The user enters to the Index page. 2. Select login link and enter email and password and click login button. 3. The system checks for validity of email, account type and password 4. entered. 5. The System displays Home page if email and password is correct. 6. Use case ends.
Alternative course of action	AL1 If users enter invalid email or password display error message like "you entered invalid email or password go to basic flow of 2. password.
Post Condition	The users logged in to the main page.

Register Player Use case Description

Table 6:Register Player Use case Description

Use case Name	Register Player Information
Description	This use case allows the Secretary to register a new player's information into the system so that the player can be managed and tracked for future activities.
Actors	Secretary
Pre-condition	The actor (Secretary) must be logged into the system. Player details (name, age, position, etc.) are available.
Basic course of action	1. The Secretary logs into the system. 2. The Secretary navigates to the "Player Management" section. 3. The Secretary clicks on the "Register Player" option. 4. The system displays the player registration form. 5. The Secretary fills in all required fields (name, age, position, team, contact details, etc.). 6. The Secretary clicks the "Save/Register" button. 7. The system validates the input and saves the player information in the database. 8. The system displays a success message confirming registration. 9. Use Case Ends.
Alternative course of action	**A1:** If the entered information is incomplete or invalid, the system displays an error message and prompts the Secretary to re-fill the form (return to Step 4). **A2:** If the player already exists in the database, the system displays a warning and stops the registration process.
Post Condition	A new player is successfully registered in the system. The player information is stored and available for future management (e.g., assigning to matches, tracking performance).

Update Player Use case Description

Table 7:Update Player Use case Description

Use case Name	Update Player Information
Description	This use case allows the Secretary to update or modify an existing player's information in the system, ensuring that the records remain accurate and up to date.
Actors	Secretary
Pre-condition	The actor (Secretary) must be logged into the system. Player record must already exist in the database.
Basic course of action	1. The Secretary logs into the system. 2. The Secretary navigates to the "Player Management" section. 3. The Secretary selects a player from the list of registered players. 4. The system displays the existing player details in an editable form. 5. The Secretary updates the required fields (e.g., name, position, team, contact, etc.). 6. The Secretary clicks the "Update/Save" button. 7. The system validates the new information. 8. The system updates the player record in the database. 9. The system displays a success message confirming the update. 10. **Use Case Ends.**
Alternative course of action	**A1:** If the Secretary leaves required fields empty, the system prompts an error and returns to Step 4. **A2:** If the Secretary tries to update a non-existent player, the system displays "Player not found" and returns to Step 2.
Post Condition	The player's information in the system is successfully updated. Updated information is immediately available for future operations (e.g., match assignment, reporting).

Approve Player Registration Use case Description

Table 8:Approve Player Registration Use case Description

Use case Name	Approve Player Registration
Description	This use case allows the Technical Director to review and approve or reject newly registered players.
Actors	Technical Director
Pre-condition	• The Technical Director must be logged into the system. • The player's registration request must already exist in the system and be pending approval.
Basic course of action	1. The Technical Director logs into the system. 2. The Technical Director navigates to the "Pending Player Registrations" section. 3. The system displays a list of players awaiting approval. 4. The Technical Director selects a player from the list to review details. 5. The system displays the player's registration information. 6. The Technical Director clicks the **"Approve"** button. 7. The system updates the player's status to "Approved" and saves the change in the database. 8. The system displays a confirmation message. 9. **Use Case Ends.**
Alternative course of action	**A1:** If the Technical Director identifies incorrect or incomplete information, they may click the **"Reject"** button instead of "Approve." The system updates the status to "Rejected" and notifies the Secretary to re-check the registration. **A2:** If no pending registrations exist, the system displays "No pending player registrations."
Post Condition	• The player's registration status is updated (either approved or rejected). • Approved players are now eligible for assignment to matches and other activities.

Register Match result Use case Description

Table 9:Register Match result Use case Description

Use case Name	Register Match Result
Description	This use case allows the Secretary to record the results of a completed match by entering goals, scores, and other performance details.
Actors	Secretary
Pre-condition	The Secretary must be logged into the system. The match must already exist in the system and must have been played.
Basic course of action	1. The Secretary logs into the system. 2. The Secretary navigates to the "Match Management" section. 3. The Secretary selects a completed match from the list. 4. The system displays the match details and a result entry form. 5. The Secretary enters the result details (final score, goal scorers, cards, and other statistics if required). 6. The Secretary clicks the **"Save/Submit"** button. 7. The system validates the input. 8. The system stores the match result in the database. 9. The system displays a success message confirming the result registration. 10. **Use Case Ends.**
Alternative course of action	**A1:** If any required result field is missing (e.g., score not entered), the system displays an error and returns to Step 4. **A2:** If the match has already been assigned a result, the system notifies the Secretary and prevents duplicate entry unless explicitly updated.
Post Condition	The match result is successfully registered in the system. The result is available for reporting, standings, statistics, and historical records.

View Match result Use case Description

Table 10:View Match result Use case Description

Use case Name	View Match Result
Description	This use case allows all actors to view the results of completed matches. Results may include final scores, goal scorers, cards, and other match statistics.
Actors	All Actors
Pre-condition	The actor must be logged into the system (if login is required). Match results must already exist in the system.
Basic course of action	1. The actor logs into the system (if required). 2. The actor navigates to the **"Match Results"** section. 3. The system displays a list of completed matches with their results. 4. The actor selects a specific match to view more detailed results. 5. The system displays the detailed match result (final score, goal scorers, cards, etc.). 6. **Use Case Ends.**
Alternative course of action	**A1:** If no results are available (e.g., no matches have been played), the system displays a message: **"No match results available."** **A2:** If the actor applies filters (e.g., by team, by date, by tournament), the system displays only the relevant results.
Post Condition	The actor successfully views the required match result(s). Results remain accessible for analysis, reports, and reference.

View Match Prediction Use case Description

Table 11:View Match Prediction Use case Description

Use case Name	View Match Prediction
Description	This use case allows actors to view predicted outcomes of upcoming matches.
Actors	All Actors
Pre-condition	• The actor must be logged into the system (if login is required). • The system must already have generated match predictions using its prediction engine or AI model.
Basic course of action	1. The actor logs into the system (if required). 2. The actor navigates to the **"Match Predictions"** section. 3. The system displays a list of upcoming matches with their predicted outcomes. 4. The actor selects a specific match to view detailed predictions. 5. The system shows details such as: • Probability of home win, away win, or draw • Expected goals for each team • Predicted key player performance • Statistical reasoning (recent form, injuries, head-to-head history) 6. **Use Case Ends.**
Alternative course of action	**A1:** If no predictions are available (e.g., due to insufficient data), the system displays: **"No prediction available for this match."**
Post Condition	• The actor successfully views the prediction of upcoming matches. • Predictions remain available for decision-making, analysis, and fan engagement.

3.3 Sequence Diagrams

UML sequence diagrams model the flow of logic within our system in a visual manner that enable us to document and validate our logic and are commonly used for both analysis and design purposes. This sequence diagrams are dynamic modeling, which focuses on identifying the behavior within our system.

Sequence Diagram for Login

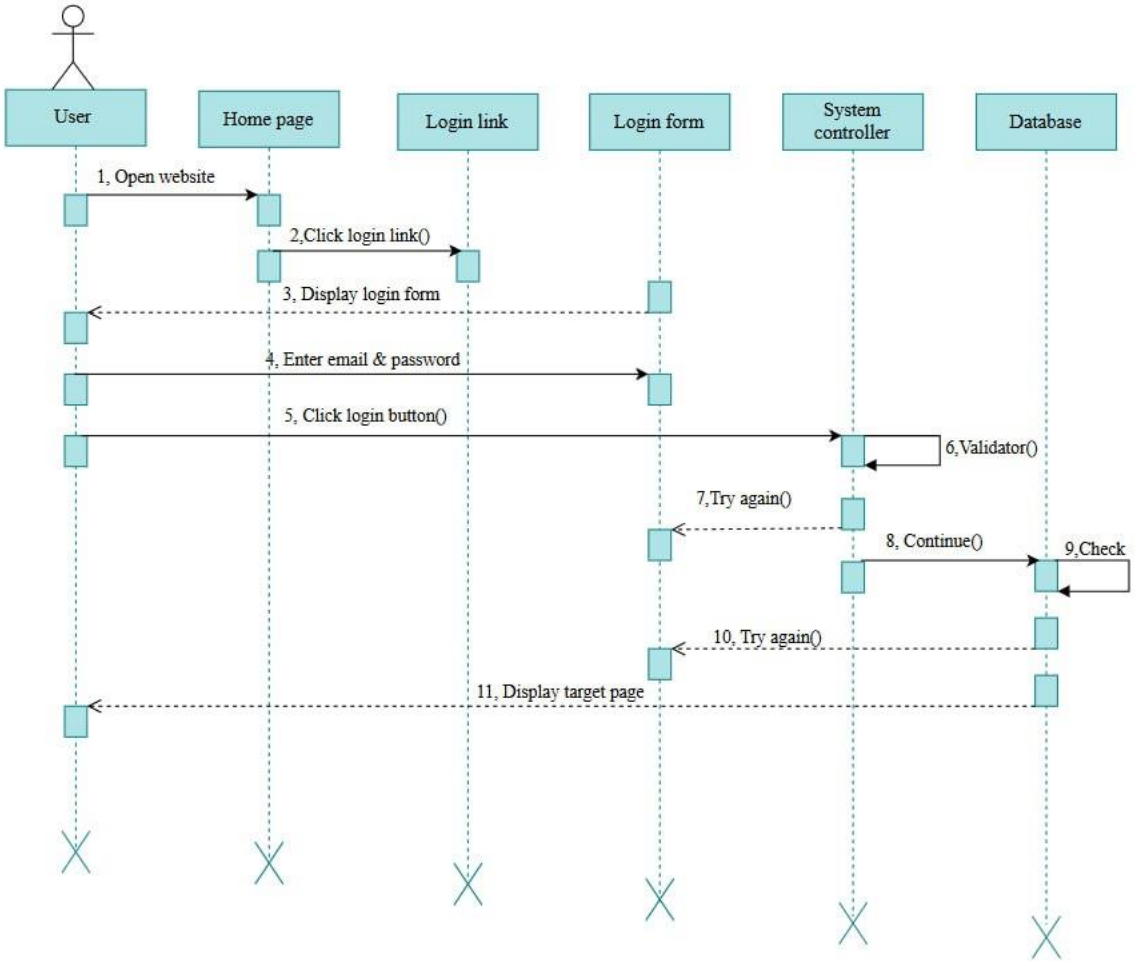


Figure 2:Sequence Diagram for Login

Sequence Diagram for Create account

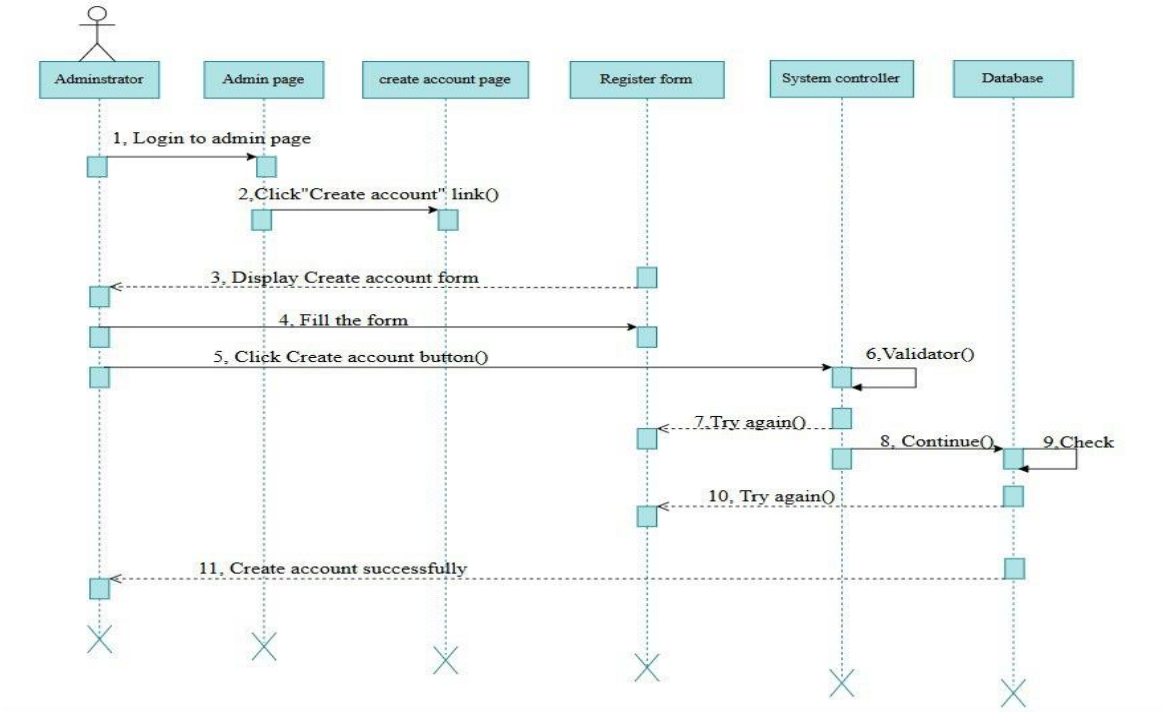


Figure 3:Sequence Diagram for Create account

Sequence Diagram for View Prediction

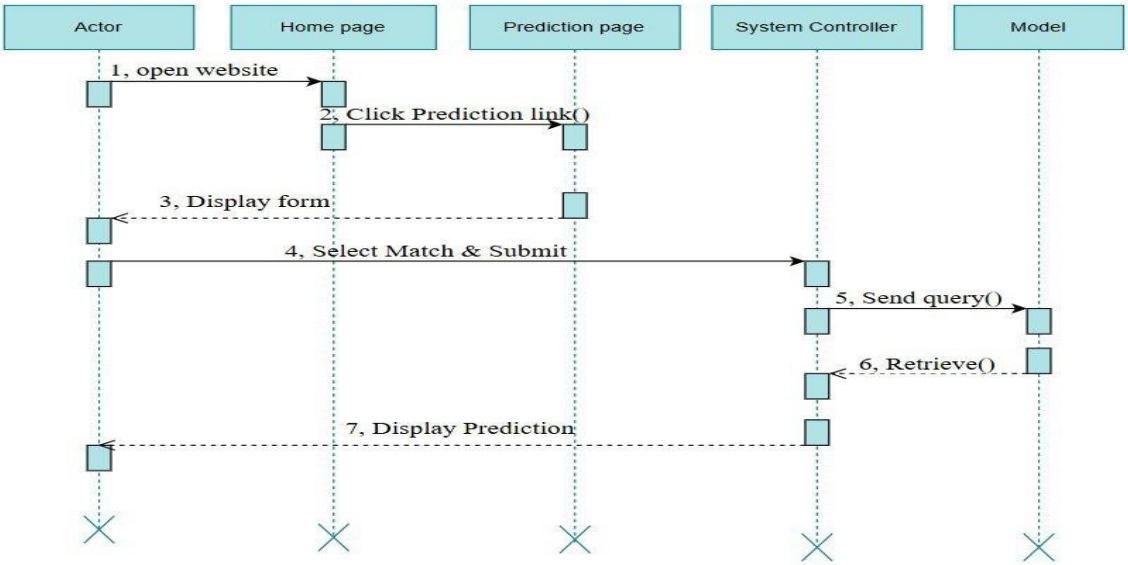


Figure 4:Sequence Diagram for View Prediction

Sequence Diagram for Register Player

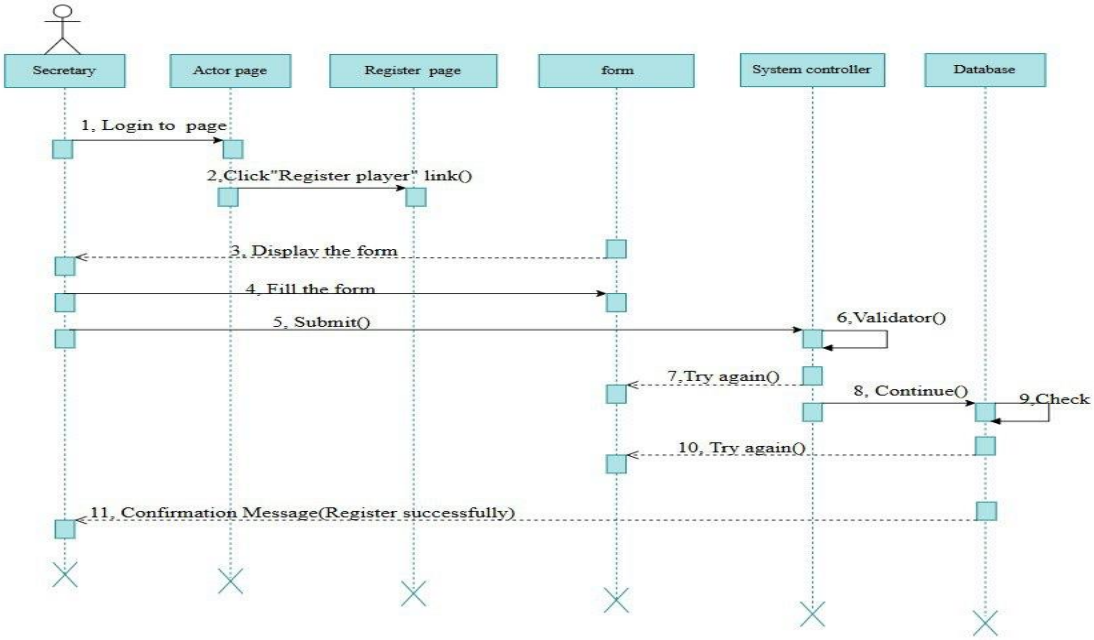


Figure 5:Sequence Diagram for Register Player

Sequence Diagram for Register Injury

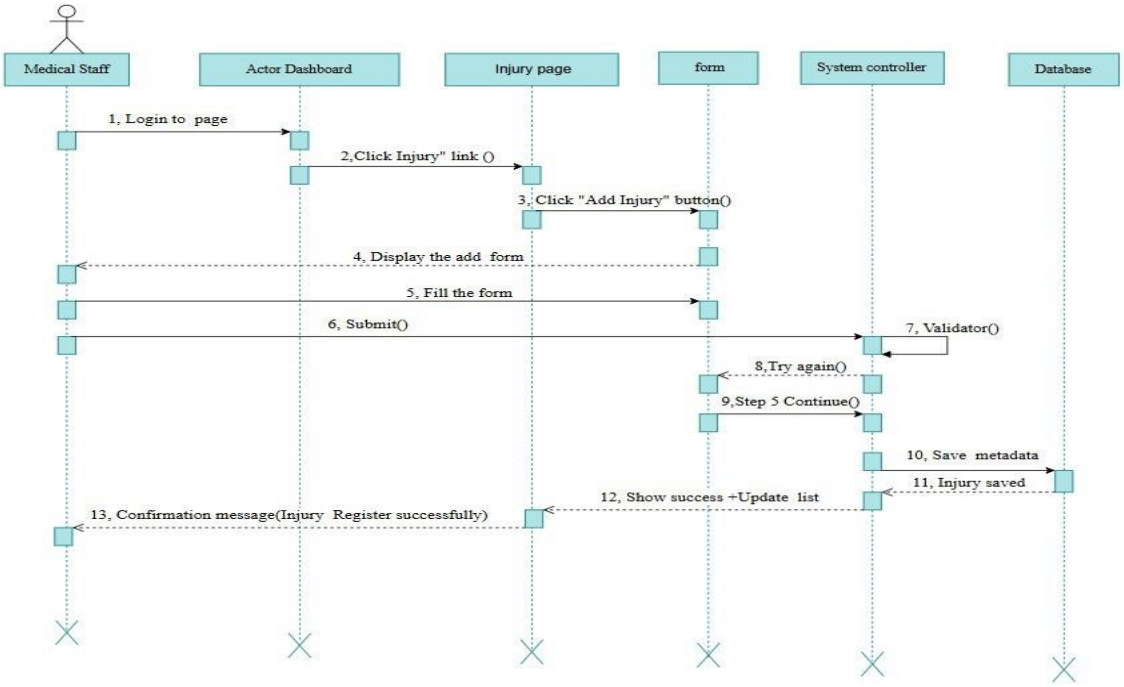


Figure 6:Sequence Diagram for Register Injury

3.4 Activity Diagrams

Activity Diagrams are used to Document the logic of a single operation /methods, a single use case or the flow of logic of a business operation. In many ways, Activity Diagrams are the object oriented Equivalent of flow charts and Data Flow Diagrams (DFD) from Structure.

An activity diagram visually presents a series of actions or flow of control in a system similar to a flowchart or a data flow diagram. Activity diagrams are often used in business process modeling. They can also describe the steps in a use case diagram. Activities modeled can be sequential and concurrent. Overall, this activity diagram provides a high-level overview of the main activities involved in the Wolaita Dicha FC management system.

Activity Diagram for login

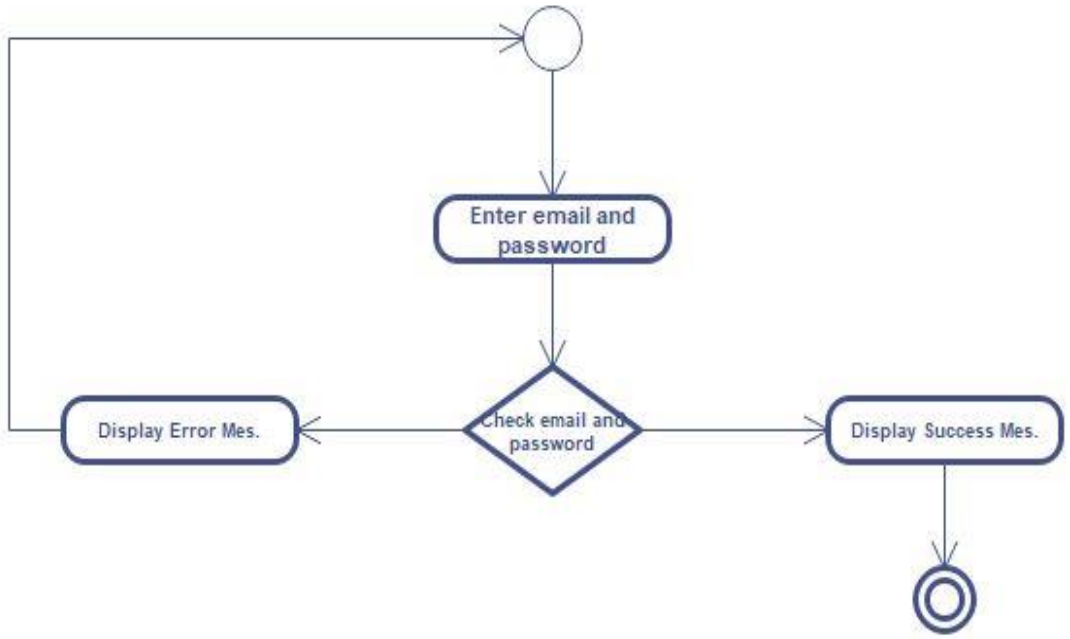


Figure 7:Activity Diagram for login

Activity Diagram for Create account

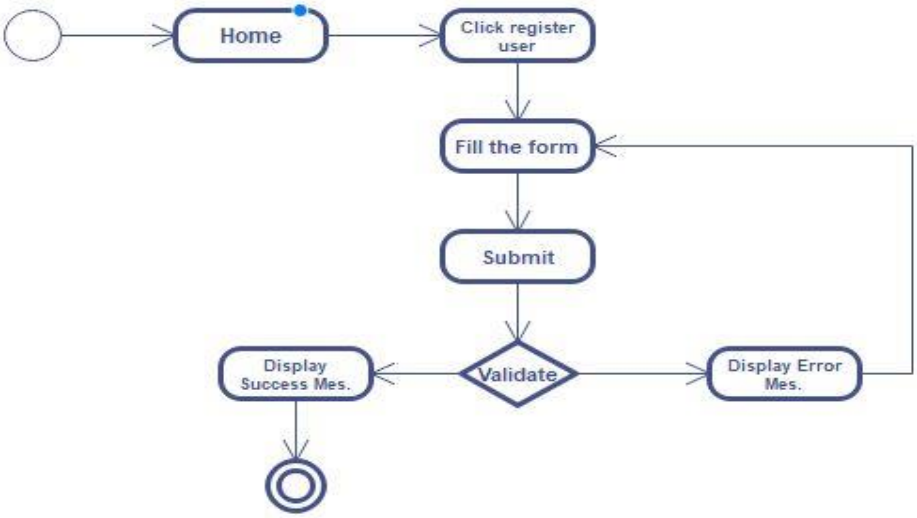


Figure 8:Activity Diagram for Create account

Activity Diagram for Register Player

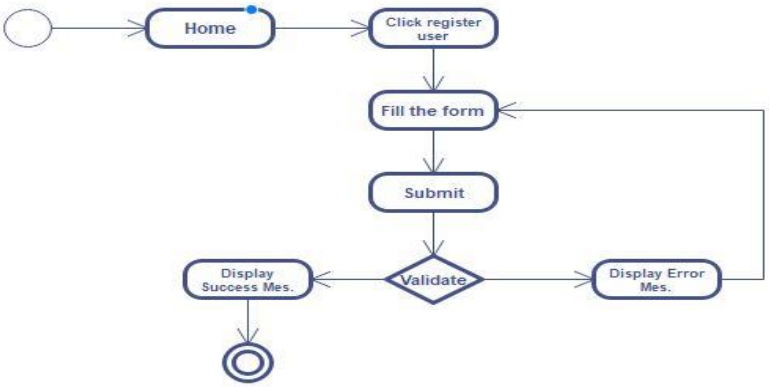


Figure 9:Activity Diagram for Register Player

Activity Diagram for View Prediction

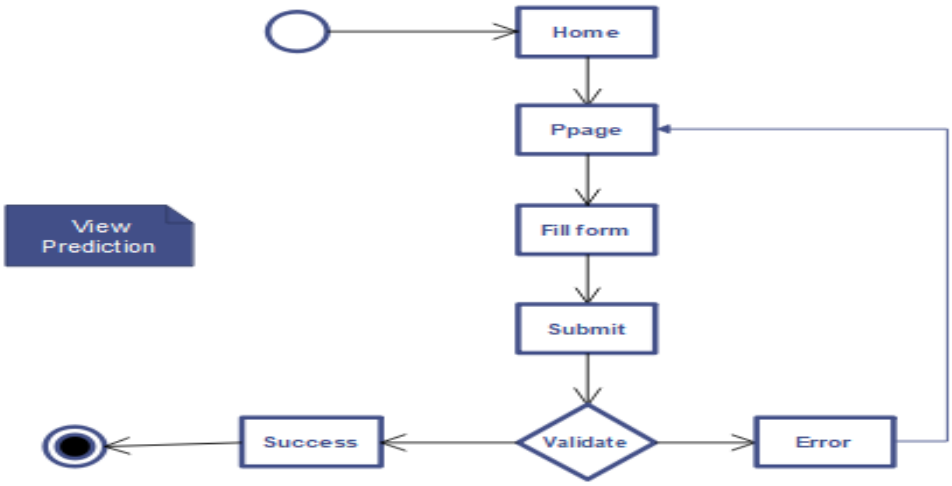


Figure 10:Activity Diagram for View Prediction

Activity Diagram for Register for Member

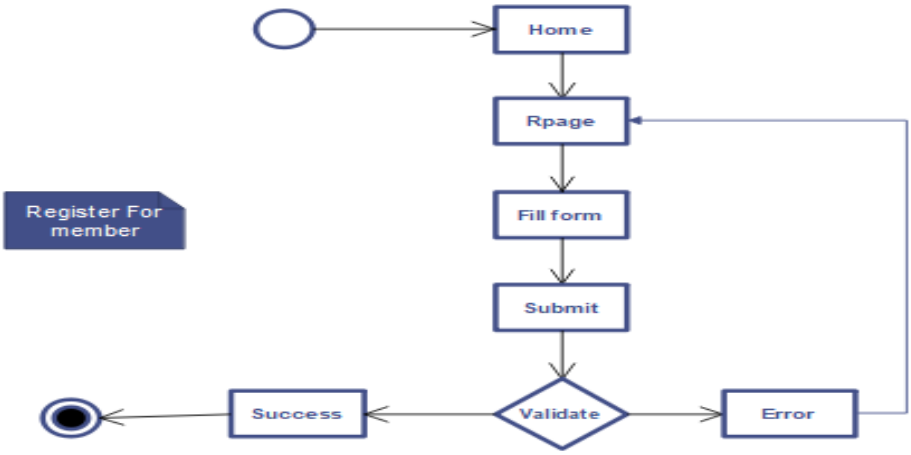


Figure 11:Activity Diagram for Register for Member

3.5 Analysis Level of Class Diagrams

Our UML analysis level class diagram is representation of meaningful concepts in a problem domain. It focuses on domain concepts (i.e., ideas, things, or objects), associations between concepts and attributes of concepts not software entities. The class diagram is the main building block of object-oriented modeling. It is used for general conceptual modeling of the structure of the application, and for detailed modeling, translating the models into programming code. Class diagrams can also be used for data modeling. This class diagram provides a high-level overview of the classes and their relationships in the Wolaita Dicha FC Management System.

Analysis Level of class diagram

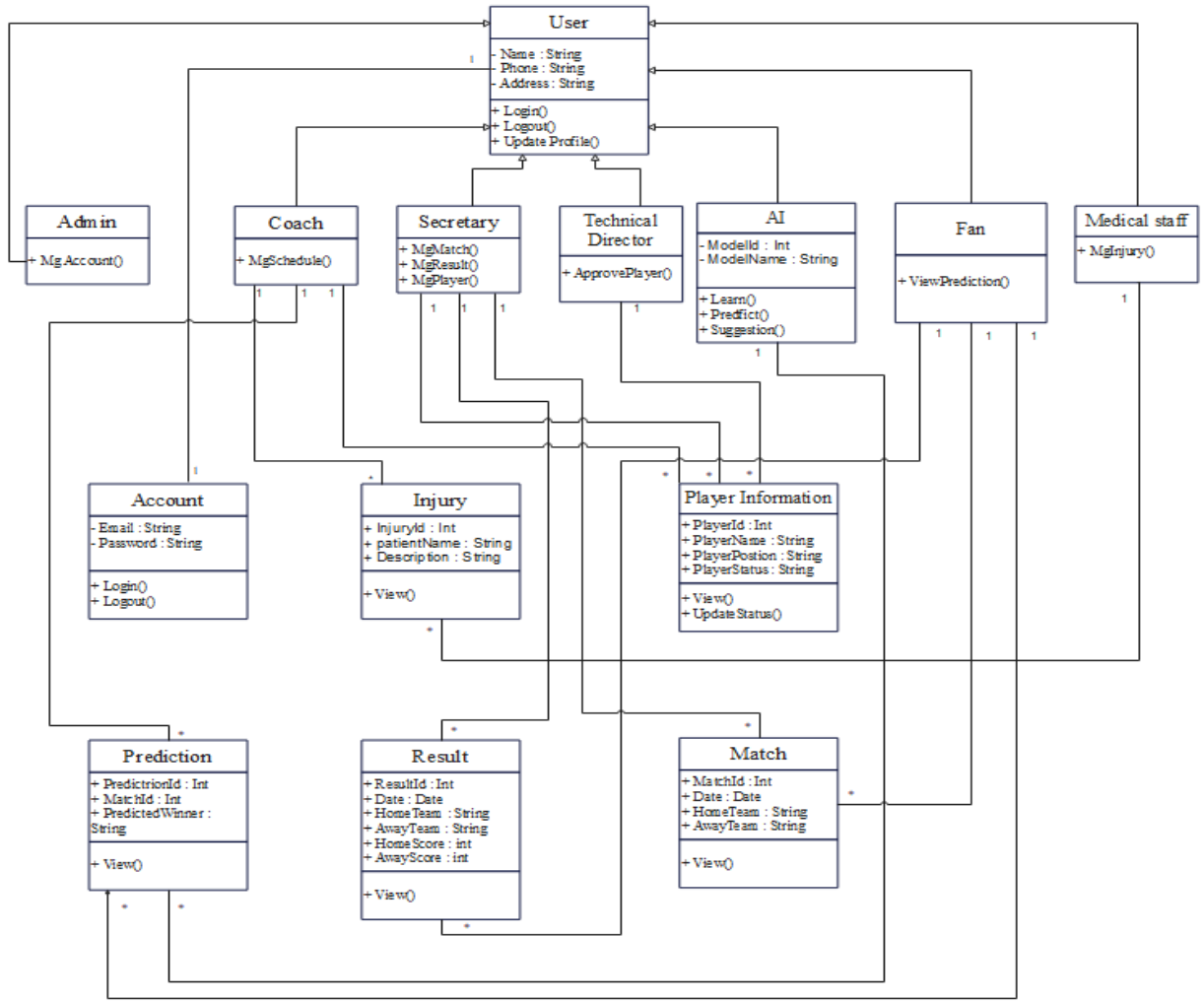


Figure 12:Analysis Level of class diagram

Chapter Four

4. Proposed Design

4.1 Introduction

System design is the process of specifying the architecture, components, modules, interfaces, and data management structures of a system so that it meets the functional and non-functional requirements of its users. It determines how data is collected, verified, processed, stored, and presented in a reliable and efficient way. A well-designed system ensures smooth interaction between users and the system while supporting scalability, security, and accuracy of operations.

The Web-Based Wolaita Dicha Football Club Management System with Match Prediction and Tactical Advice is designed to provide an integrated platform that supports both administrative management and football performance enhancement. Traditional club management systems primarily focus on handling player records, match schedules, and general reporting. However, these systems lack advanced capabilities such as predictive match outcomes and AI-driven tactical guidance, which are critical for modern football decision-making.

The proposed system not only manages administrative tasks like player registration, match scheduling, and result recording but also incorporates **AI-powered prediction and tactical analysis**. This enables coaches, technical staff, and administrators to gain insights into upcoming matches, anticipate possible outcomes, and receive tactical advice to improve team performance. By combining administrative management with predictive analytics, the system enhances decision-making, streamlines operations, and strengthens the competitive advantage of **Wolaita Dicha Football Club**.

4.2 Design Goal

Design goals describe the important system qualities and define the values against which design alternatives are evaluated. When designing a new system, the designer creates a model from user requirements, and the quality of that model determines the quality, usability, and maintainability of the final product.

The design of the Web-Based Wolaita Dicha Football Club Management System with Match Prediction and Tactical Advice is guided by a set of goals that ensure the system is reliable, intelligent, and capable of supporting both administrative management and AI-driven decision-

making. The primary design goals are as follows:

Efficient Club Management

- Provide a platform for administrators and secretaries to manage player registrations, match schedules, and match results effectively.
- Ensure that all operations (registration, updates, and approvals) are streamlined to minimize errors and delays.

Integration of Match Prediction and Tactical Advice

- AI models to analyze historical match data, player performance, and team statistics for predicting match outcomes.
- Provide tactical advice to technical directors and coaches, highlighting opponent weaknesses, key strategies, and recommended formations.

Reliability and Availability

- Guarantee high availability of the management system, especially during peak periods such as match days.
- Include backup and recovery mechanisms to ensure uninterrupted service and data protection.

4.3 Current Software Architecture

At present, football club management in Wolaita Dicha Football Club is largely handled using manual or semi-automated processes, which limits efficiency, performance analysis, and decision-making. The architecture of the existing/manual system can be described as follows:

User Interaction Layer

- Manual Registration: Player details are collected using paper forms or simple spreadsheets maintained by the secretary.
- Match Scheduling: Fixtures are planned manually and communicated verbally or through notices.
- Result Recording: Match outcomes are written on paper or stored in basic spreadsheets without centralized access.
- No Tactical Support: Coaches and staff rely on personal judgment without system-generated advice.

Communication & Notification

- Manual Communication: Information about matches, results, or player status is shared via phone calls, printed notices, or face-to-face meetings and it uses telegram channel.
- Delayed Updates: Important updates such as schedule changes or player approvals take time to reach all stakeholders.
- No Automated Alerts: Players, staff, or fans are not automatically notified of schedules, results, or predictions.

Data Management

- Paper-Based Records: Player and match data are kept in files, notebooks, or basic Excel sheets.
- Lack of Centralized Database: No unified system to manage all records; data is scattered across documents.
- Limited Historical Analysis: Past data is rarely used for meaningful performance evaluation or predictions.

4.4 Proposed Software Architecture

The proposed Web-Based Wolaita Dicha Football Club Management System with Match Prediction and Tactical Advice is designed to automate administrative tasks, centralize data management, and integrate AI-driven prediction and tactical support. This architecture ensures real-time access, improved communication, and intelligent decision-making. It can be described as follows:

Web Browser

These include Mozilla Firefox, Google Chrome, Opera Mini, Internet Explorer, and some others browsers. These are used to run the system in the computer system. These operating systems can be installed in the client computers and therefore can be accessed as much as needed.

- Web-Based Interface: Accessible through browsers on desktop and mobile devices.
- Automated Forms: Supports online player registration, match scheduling, and result entry.

- Interactive Analytics: Provides visual dashboards with match predictions, performance graphs, and tactical advice.

Application & Processing Layer/ Web Server

It is the heart of system. It has a lot of responsibilities to perform like to communicate with clients and database. Since the system is network-connected, it uses HTTPs to send and receive data. HTTPs Protocol used for communication between two devices/machines over the network to interact with each other.

- Player Management Module: Handles registration, updates, and approval of players.
- Match Management Module: Automates scheduling, result recording, and match history tracking.
- AI Prediction Engine: Uses machine learning models trained on historical data to predict match outcomes (win/loss/draw probabilities).
- Tactical Advice Module: Generates insights on team strengths, weaknesses, and opponent analysis to guide coaches.
- Authentication & Authorization: Implements secure login with role-based access.

MySQL Database/ Data Management Layer: - Is the database software deals with the storing and managing the data based on the request from the web server. It will write new records to the storage and will retrieve and deliver to the web server when needed. MySQL is one of the most popular database technologies in the world. Known for being secure and scalable, it is the go-to solution for businesses of all sizes.

- Centralized Database: Stores player details, match schedules, results, and AI outputs.
- Historical Data Storage: Maintains past records for performance analysis and continuous AI training.
- Data Security & Backup: Implements encryption, access control, and backup mechanisms to ensure data integrity and availability.
- Dataset: - is a structured collection of data, usually organized in a way that makes it easy to access, manage, and analyse.

AI & Analytics Layer

- Predictive Analysis: Forecasts possible match outcomes using historical and real-time data.
- Tactical Insights: Recommends formations, strategies, and adjustments based on opponent weaknesses and team performance trends. [2]
- Data Visualization**:** Provides charts and reports for decision-making by the technical director and

Proposed System Architecture

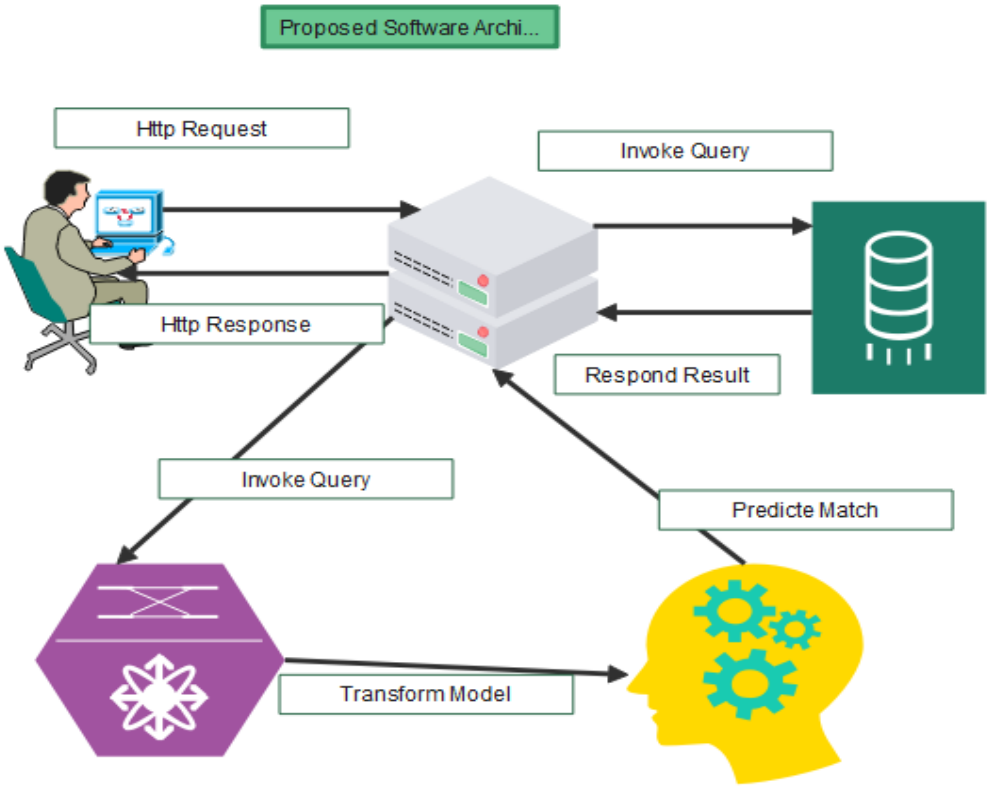


Figure 13:Proposed System Architecture

4.5 Overview

The Wolaita Dicha Football Club Management System with Match Prediction and Tactical Advice is an intelligent, centralized platform designed to streamline football club operations while enhancing team performance through data-driven insights. The system combines traditional club management functions such as player records, match results, and training schedule with advanced predictive analytics to forecast match outcomes and provide tactical recommendations for coaching staff.

There are three main perspectives of system design:

Architectural Design

The architectural design focuses on the overall structure and behavior of the system. It defines how different components such as user interfaces, data management, analytics modules, and reporting tools interact to provide a cohesive and efficient platform. This layer ensures that the system is scalable, maintainable, and capable of integrating AI-based predictive models effectively.

Logical design

The logical design provides an abstract representation of the system's data flows, inputs, and outputs. It typically involves modeling tools such as entity-relationship diagrams (ERDs) to define the relationships between players, matches, training sessions, and performance metrics. This design ensures that data is logically organized, enabling accurate analysis and prediction.

Physical design

Physical design deals with the implementation details of the system:

- **User Interface Design** focuses on how coaches, managers, and staff interact with the system to enter data, view match predictions, and receive tactical advice.
- **Data Design** defines how player statistics, match records, and historical performance data are represented, stored, and retrieved efficiently.
- **Process Design** ensures the proper flow of data through the system, including validation, security, and transformation of information as it moves between input, analysis, and reporting modules.

4.6 Sub System Decomposition

Subsystem decomposition is the process of breaking down a large, complex system into smaller, more manageable parts called subsystems. The main purpose of this process is to simplify the overall design, development, and maintenance of the system.

Sub system Decomposition

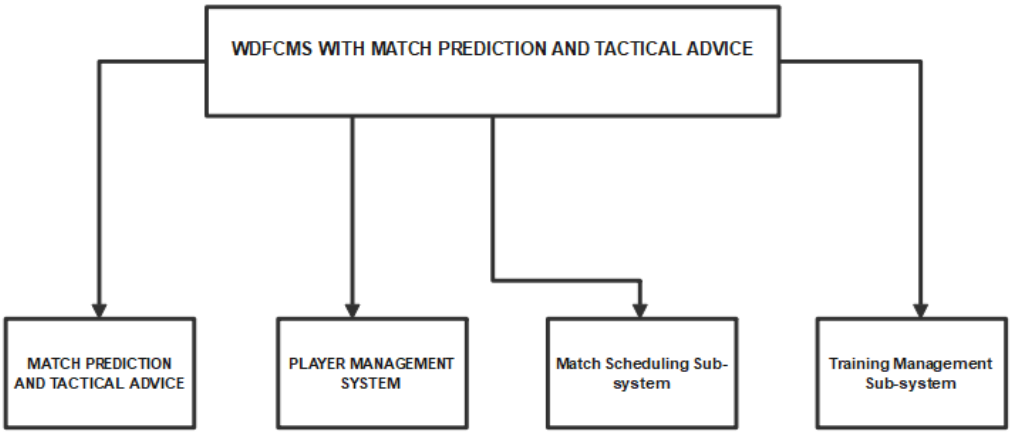


Figure 14:Sub System Decomposition

4.7 Hardware Software Mapping

Deployment diagram depicts a static view of the run-time configuration of processing nodes and the components that run on those nodes. In other words, deployment diagrams show the hardware for our system, the software installed on that hardware, and the middleware used to connect the disparate machines to one another. Accordingly; WDFC management system deployment diagram consist the system software installed on electronic devices with browsers acting as middleware to connect client side to the server side.

Hardware and Software Mapping

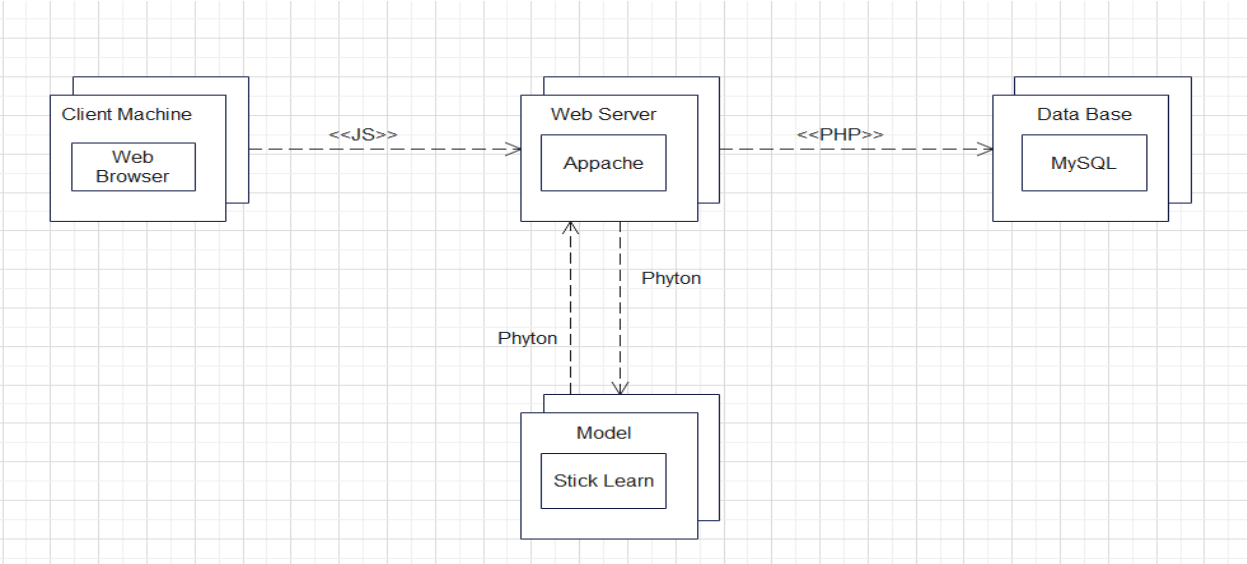


Figure 15:Hardware and Software Mapping

4.8 Deployment Diagram

Deployment diagram depicts a static view of the run-time configuration of processing nodes and the components that run on those nodes. In other words, deployment diagrams show the hardware for our system, the software installed on that hardware, and the middleware used to connect the disparate machines to one another.

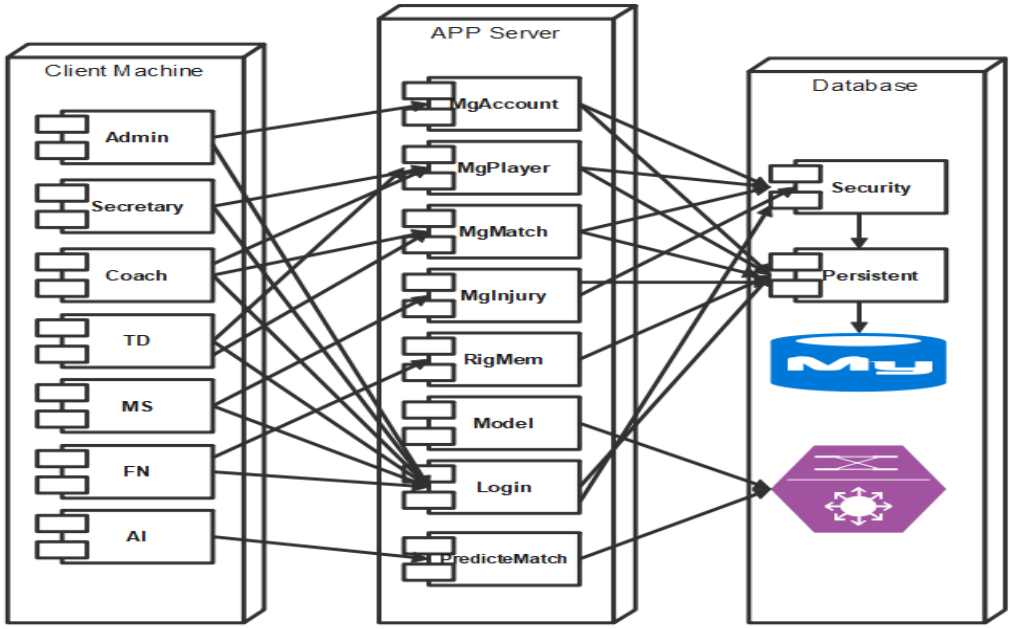


Figure 16:Deployment Diagram

4.9 Persistence Data Management

Persistence data management in the Wolaita Dicha FC Management System ensures secure, reliable storage and retrieval of all club-related data including players, matches, training, contracts, medical records, and memberships. The system uses MySQL as the primary database due to its flexibility, scalability, and ability to handle document -oriented data.

Core entities include Player, Team, Match, Training Schedule, Contract, Medical Report, User, and Membership. Each role—Administrator, Coach, Technical Director, Medical Staff, Secretary, Fan, and AI Model—interacts with the data according to permissions defined by the s ystem.

Daily incremental and weekly full backups are implemented to ensure data safety and recovery. Player and contract information managed by Administrators, Coaches, and Technical Directors is retained permanently for historical accuracy. Medical Staff records and updates injury and fitness reports linked with player performance, while Secretaries handle match registrations. Fans' membership data and payments are securely stored and retained for compliance and operational needs. AI Models access anonymized datasets to generate predictions and tactical advice without compromising live data integrity.

Data integrity is maintained through schema validation, unique indexes, and transactional operations. Role-based access controls enforce security and limit data manipulation according to user responsibilities. Localization features such as UTF-8 encoding and geographic metadata tagging support multi-language content and regional operations. All critical operations are logged for auditing and compliance.

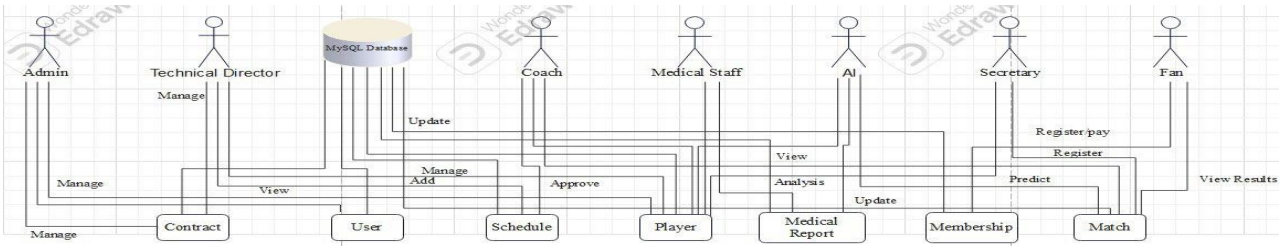


Figure 17:persistence diagram

4.10 Access Control and Security

Access control is the process of restricting access to a system, resource, or piece of information. In software development, it's a critical component of security that ensures only authorized users can perform specific actions, view certain data, or utilize particular features. This is often achieved through a combination of authentication and authorization.

Table 12:Access Control and Security

Actors	Administrator	Secretary	Coach	Fan	MS	TD	AI
Register Users	✓						
Update account	✓						
Search user	✓						
Activate user	✓						
Deactivate user	✓						
Forget password	✓						
Change password	✓						
Update profiles	✓						
Login	✓	✓	✓	✓	✓	✓	
Register players		✓					
Update player		✓					
Search player		✓					
Register matches		✓					
Update matches		✓					
Search matches		✓	✓	✓	✓	✓	
Register results		✓					
Update results		✓					
ASQ ai		✓	✓	✓	✓	✓	
View matches		✓	✓	✓	✓	✓	
predicate Matches							✓

Security Mechanism:

User authentication employs a password hashing technique using the **SHA-1 algorithm**, converting each password into a 160-bit hash before database storage. This ensures that plaintext credentials are never stored or transmitted. Although SHA-1 is used here for demonstration, stronger algorithms such as **SHA-256** or **bcrypt** are recommended for production environments to achieve higher cryptographic strength.

4.11 Global Software Control

Due to its nature as a web-based system, the Wolaita Dicha Football Club Management System will be event-driven. When a user interacts with a web page, they will have the option of clicking on various links or buttons. When an action is performed, a corresponding function in the system management subsystem will be invoked. Functions within a subsystem may call other functions within the same subsystem, or they may communicate with other subsystems through the interfaces provided.

- The coach initiates the Match Prediction and Tactical Advice Subsystem by logging in, allowing access to predicted match outcomes, tactical recommendations, and player performance analytics.
- The team analyst initiates the Player Performance Management Subsystem by logging in to update player statistics, injury reports, and training progress.
- The club administrator initiates the Account and User Management Subsystem by logging in, managing access permissions, team rosters, and system configuration.
- Fans and supporters initiate the Fan Engagement Subsystem by logging in to view match results, upcoming fixtures, and club news.

4.12 Boundary Condition

Startup and Shutdown Condition

The Wolaita Dicha Football Club Management System does not have any dedicated startup or shutdown procedures. Since it is a web-based system, startup and shutdown are managed by the web server and hosting environment. The system is designed to be accessible 24 hours a day, 7 days a week, and 365 days a year, except during planned maintenance or updates. If the web server and host are running, the Wolaita Dicha Football Club Management System is fully operational.

Initialization

The system does not require specific initialization when starting up. When the website is created for the first time, some essential persistent data such as player profiles, team rosters, and historical match records needs to be added to the database. However, these operations can be performed at any time. If the web server is restarted or the system experiences downtime, it is not necessary to re-enter previously stored data, since all critical information is already persisted in the database.

4.13 Sub System Service

The Wolaita Dicha Football Club Management System is composed of multiple subsystems, each responsible for a specific set of functionalities. These subsystems interact with each other under the supervision of the **Global Software Control**, ensuring smooth operation and data consistency.

Interaction between Subsystems:

Match Prediction & Tactical Advice Subsystem ↔ Player Performance Management Subsystem

- The tactical subsystem relies on updated player statistics, injury reports, and training performance from the player performance subsystem to generate accurate predictions and tactical suggestions.
- Any changes in player availability or fitness are immediately communicated to the tactical subsystem.

Match Prediction & Tactical Advice Subsystem ↔ Account & User Management Subsystem

- Access control ensures that only authorized personnel (coaches, analysts) can view or modify predictions and tactical advice.
- The user management subsystem verifies login credentials and roles before granting access to sensitive data.

4.14 User Interface Modeling

The Wolaita Dicha Football Club Management System is designed with a web-based user interface that is intuitive, responsive, and role-specific. The UI allows different users coaches, analysts, administrators, and fans—to access the system's features efficiently.

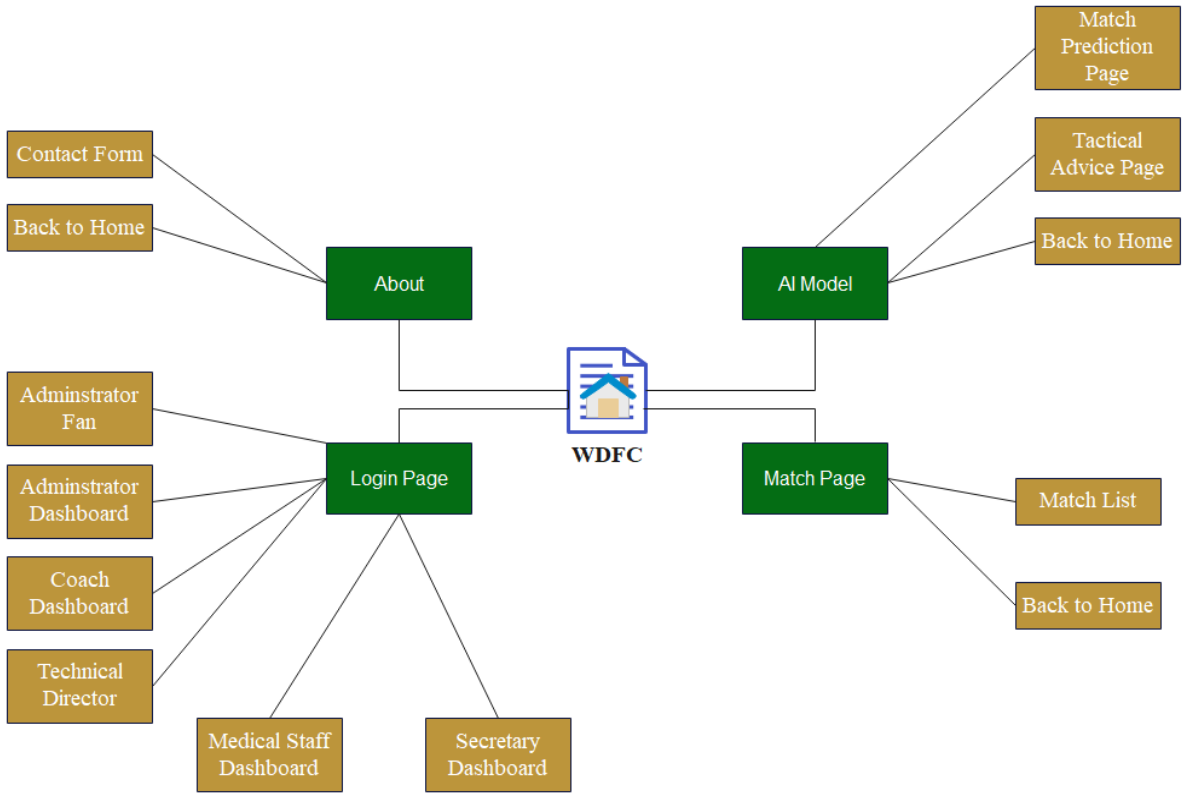


Figure 18:User Interface

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